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RESEARCH ARTICLE

GenAI-Based Jamming and Spoofing Attacks on UAVs

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ABSTRACT Recently, aerial vehicles have been more connected than ever, where there are many types of the vehicles. Uncrewed Aerial Vehicles (UAVs) operate on various environments with different technologies that are subject to many attacks. Creating effective intrusion detection systems against such attacks has been a significant challenge since there is a lack of sufficient attack data that can be used to design an intrusion detection system with advanced computing algorithms. In this research, we propose a novel framework to create attacks data for UAVs by using generative artificial intelligence algorithms. We use Variational Autoencoder, Gaussian Copula, Denoising Diffusion Probabilistic Model (DDPM), and Conditional Tabular Generative Adversarial Network to create synthetic attack data. Specifically, jamming and spoofing attacks on UAVs are generated to fool intrusion detection systems that may be implemented on UAVs. Experimental evaluations show that synthetically generated attack data reduces the accuracy of intrusion detections if the system was trained with inadequate attack data. Additionally, analysis results show that DDPM emerged as the most effective model for generating attack data, leading to F1 score reductions of 21% for jamming and 28% for spoofing attacks. This research highlights the need for more robust and adaptive intrusion detection systems that can be created with synthetic data. Thus, sustainable computing systems on UAVs will be achieved.

INDEX TERMS Cyber security, generative artificial intelligence, intrusion detection system, synthetic data, uncrewed aerial vehicles.

I. INTRODUCTION

Uncrewed Aerial Vehicles (UAVs) are aerial vehicles that operate autonomously under remote human control or via onboard computing systems. The rapid advancement and the commercialization of communication networks and artificial intelligence have made UAVs increasingly versatile with their ability to operate in difficult and inaccessible environments. UAVs have become an integral part of a wide range of civil and military applications since they provide high adaptability and cost-effectiveness. The vehicles use many technologies that require interconnected networks of sensors, complex software, data collection, flight control, mission execution. Moreover, the operational environment is evolving dynamically. These conditions introduce different vulnerabilities on UAVs, which expands the threat surface considerably. Specifically, UAVs communication systems are the most vulnerable part of UAVs against cyber attacks. Adversaries often attempt to disrupt UAV operations by intercepting their communications, compromising the confidentiality, integrity, and availability of data. For instance, jamming attacks like Denial-of-Service (DoS) and Global Positioning System (GPS) spoofing, replay, routing, eavesdropping, and false data injection are such examples, which disrupt UAV availability by affecting communications [1], [2], [3].

Jamming and GPS spoofing attacks have been used to disrupt wireless sensor networks, such as MavLink protocol and severely affect UAV communications. Jamming attacks are intentionally used to disrupt or block communication signals. Such attacks directly compromise system availability and pose serious challenges to UAV security [4]. GPS spoofing attacks, which alter the trajectory of autonomous

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UAVs, represent another critical threat to UAV systems [5], [6]. The most critical vulnerabilities include GPS spoofing and GPS jamming attacks, which disrupt communications and undermine the availability of UAVs. In this research, we focus on GPS jamming and GPS spoofing attacks on UAVs.

Recent real-world case studies underscore the urgent need for robust Intrusion Detection Systems (IDS) capable of mitigating GPS spoofing and jamming attacks on UAVs. For instance, Pocock (2012) reported a UAV crash in South Korea directly linked to GPS jamming, highlighting how interference in GPS signals can lead to total mission failure or even physical destruction of the vehicle [7]. Similarly, Shafique et al. illustrated that a spoofed UAV trajectory can cause the drone to deviate significantly from its intended flight path, enabling adversaries to hijack or mislead the drone without the operator's knowledge [8].

For example, in 2011, Iranian forces allegedly hijacked a U.S. RQ-170 Sentinel drone using GPS spoofing techniques, forcing it to land in hostile territory by deceiving its navigation system with counterfeit signals [9]. Similar concerns resurfaced in 2019 when a commercial drone operating in the Middle East was reportedly compromised through GPS spoofing, deviating from its intended path and landing in an unauthorized zone [10]. These events highlight the potential consequences of spoofing attacks, including the disruption of military, commercial, and emergency services operations.

In civilian contexts, vulnerabilities in small UAVs have also been demonstrated using low-cost Software-Defined Radios (SDRs), which allow attackers to inject counterfeit GPS signals and manipulate navigation systems remotely. Furthermore, GPS spoofing was shown to result in incorrect delivery locations for logistics drones and misguided targeting in military UAVs, posing severe threats to both safety and security [11].

According to real-world consequences, our proposed framework becomes even more critical. By generating synthetic GPS spoofing and jamming attack data using advanced generative models, we provide a scalable and ethically sound solution for training IDS without exposing sensitive infrastructure or incurring the cost and risk of live attack replication. This research directly contributes to enhancing UAV resilience in operationally realistic scenarios, as reflected by the alignment of our synthetic attacks with real-world vulnerabilities.

Security measures have been proposed to mitigate jamming and GPS spoofing threats, such as encryption, authentication protocols, and IDS. Specifically, IDS plays an important role in protecting UAV systems by monitoring data traffic and identifying unauthorized or malicious activities. Machine learning-based IDSs have shown promising results in addressing signal spoofing and jamming attacks by analyzing data from UAV flight and sensor readings [12].

Existing IDS solutions for UAVs have several limitations, which prevent to have secure UAV systems. For example, many IDS systems produce excessive false alarms, leading to unnecessary operational disruptions. IDS systems may struggle to provide a real-time threat detection. On the other hand, UAV threats evolve rapidly and existing IDS systems often fail to adapt to new and unknown attack vectors. UAVs typically operate with a limited power supply and a communication bandwidth, which may restrict the deployment of advanced IDS algorithms. Moreover, UAVs have constrained computational and storage capabilities that limit IDSs to implement complex algorithms onboard.

Training IDS systems, particularly those based on machine learning, requires extensive data that include diverse attack scenarios. Such data are often unavailable or incomplete for UAV environments. Training on imbalanced datasets with significantly fewer attack samples than normal operations leads to a poor detection accuracy for rare threats. These highlight the need for more robust, adaptable, and resource-efficient systems that may address the diverse and evolving threat landscape. The challenge in this research is the lack of attack data about UAVs for IDSs. We propose a novel framework to generate synthetic attack data about UAVs by using real data using generative AI. Specifically, we introduce a Generative Artificial Intelligence (GenAI) framework to generate attack data that are used to enhance an IDS performance to protect UAV security. Unlike traditional approaches, the proposed framework may learn different data distributions within UAV systems and produce both realistic attacks and more malicious data. While prior studies have utilized GANs to generate adversarial data primarily for evasion or deception testing of IDS models, our work introduces a multi-model synthetic attack generation framework specifically designed for UAV environments. Unlike conventional GAN-based approaches that focus on a single generative model or a narrow attack scope, our framework leverages multiple generative AI models to generate diverse synthetic jamming and spoofing attack patterns. The main contributions of this research are as follows.

- Utilized multiple generative models—Variational Autoencoder (VAE), Gaussian Copula, Conditional Tabular GAN (CTGAN), and Denoising Diffusion Probabilistic Model (DDPM)—to create diverse synthetic jamming and spoofing attack data in UAV systems.
- Developed to quantify how different generative models affect IDS robustness in UAV security.
- Proposed a multidimensional evaluation framework that combines statistical similarity metrics (e.g., Kullback–Leibler (KL) Divergence, Wasserstein Distance, Kolmogorov-Smirnov (KS) Test) with downstream IDS classification performance (accuracy, recall, precision, F1-score) to holistically assess the quality and impact of synthetic attack data.
- Introduced a cross-testing strategy using different training sets (real, synthetic, hybrid) to measure the generalizability and robustness of IDS models against unseen attack patterns.

The rest of the paper is organized as follows. Section II contains a literature review about attacks on UAVs, existing security mechanisms. Section III introduces the proposed framework. Section IV describes attack scenarios. Section V is about the experimental setup. Section VI contains experimental results. Section VII contains the discussion. The last section is devoted to the conclusion.

II. UAV AND SECURITY

UAV systems contains significant security threats primarily due to their dependence on wireless communication channels, uncontrolled environments, limited resources, dynamic topologies, and their collaboration with other UAVs [13]. This section overviews the state of the art about UAV security.

A. UAV THREATS

UAV networks are vulnerable to a range of attacks that target their communication and routing systems. Attackers may target network traffic, disrupt routing operations, or introduce malicious nodes into the network. The literature extensively documents various threats to UAV networks, emphasizing their impact on the privacy, integrity, and availability of these systems. For example, [3] provides an overview of cyber-attacks on UAVs, while [14] highlights the critical importance of securing communication links between ground control stations and UAVs.

Numerous researches focus on identifying security threats in UAV communications and creating countermeasures. In [15], the research provides a foundation for understanding key challenges in uncrewed systems, including privacy and security issues. Common examples of attacks include the interception of critical communications by adversaries and GPS spoofing, which manipulates the flight paths of autonomous UAVs [5], [6], [16]. Attackers can exploit vulnerabilities in communication protocols through various strategies [17].

Jamming attacks pose a critical threat to UAV operations by disrupting communication signals between UAVs and Ground Control Stations (GCS). These attacks typically involve the transmission of high-power radio frequency (RF) signals, which overpower legitimate communication channels and degrade system performance. Such disruptions can lead to the loss of control, forcing UAVs to either land prematurely or crash. Jamming also reduces the signal-to-noise ratio (SNR), increases packet loss, and severely impairs navigation systems [18].

To address these risks, several countermeasures have been proposed, including Frequency Hopping Spread Spectrum (FHSS), adaptive filtering, and machine learning-based detection techniques. For instance, Aldosari et al. [19] introduced a machine learning approach that detects UAV jamming attacks and classifies network nodes based on received power and distance ratios. Their work demonstrates that high-accuracy jamming detection (up to 99.7% with KNN) is feasible using distributed single-node observers, significantly reducing overhead compared to centralized detection. The integration of Deep Neural Networks (DNNs), eXtreme Gradient Boosting (XGB), and distributed detection techniques has also proven effective in improving jamming detection and response capabilities [20].

Spoofing attacks are another significant threat where adversaries manipulate GPS signals to mislead UAVs about their locations and trajectories. This attack may cause UAVs to deviate from their intended flight paths, potentially leading to catastrophic consequences, such as collisions or diversion to hostile territories.

Recent researches use machine learning and ensemble methods, such as AdaBoost-Convolutional Neural Network (CNN) and stacked ensemble models, to enhance spoofing detection accuracy even with limited samples [20], [21]. The integration of artificial intelligence and machine learning for real-time threat detections and responses remain under-explored [18]. Furthermore, the scalability and the cost-effectiveness of existing countermeasures require validation in real-world scenarios. The complexity of GPS spoofing and jamming techniques highlights the need for adaptive and robust counter-strategies [18], [20], [21]. In the literature, there is a lack of research that addresses UAV threats and their countermeasures with machine learning to ensure their safe and reliable operation.

The use of machine learning-based IDS has recently become a popular approach for securing UAV networks. For example, a machine learning-based IDS was proposed in [12] to address GPS spoofing attacks. This approach utilizes one-class support vector machines and ML autoencoder algorithms to identify anomalous behavior. Similarly, a detection method for eavesdropping attacks on UAV communication is presented in [22]. This approach employs a combination of K-means clustering and support vector machines for the analysis of data and the prediction of future threats. The method operates in two phases. Initially, both communicating parties send signals to the UAV. Subsequently, the UAV relays these signals to a third party, which identifies any deviations. Subsequently, the dataset is classified using machine learning algorithms with the objective of detecting potential attacks.

Machine learning-based IDSs are used to mitigate signal spoofing and jamming attacks. A self-learning approach with multi-class support vector machines ensures a high true positive rate in IDS as in [23]. The system incorporates a deep-Q network, a deep reinforcement learning algorithm, for dynamic route learning as a self-healing mechanism during the IDS recovery phase. The solution collects data from various sources within the UAV, including flight logs and sensor readings. A CNN approach for detecting jamming signals is proposed in [24]. This method considers weights and values of GCS and it selects a relay power element based on the bit error rate. While this algorithm effectively protects communications against jamming attacks, the random selection of relay power may increase the error rate that may result in high computational costs. In addition to these methods, a variety of machine learning security frameworks have been

developed to address a range of security challenges, including the detection and prevention of malicious drone activity and DoS attacks [25]. Recent developments indicate that federated learning methodologies may prove more effective than traditional machine learning algorithms. For example, radio frequency-based UAV authentication models utilizing IoT networks have been constructed using federated learning techniques [26].

UAVs rely on wireless communication, making them vulnerable to attacks, such as GPS spoofing, data injection, and DoS attacks. Traditional detection methods rely heavily on pre-defined parameters and they often struggle with emerging threats [27], [28]. For instance, MUVIDS system emphasizes network-level IDS but it faces scalability challenges due to its reliance on MAVLink protocol analysis [28].

Each type of attack has specific targets on UAVs. An attack compromises distinct security requirement. A comprehensive security strategy requires the integration of multiple defense techniques. Additionally, a power consumption and a battery usage must be carefully assessed to identify suitable countermeasures against such attacks. Furthermore, defense mechanisms should be continuously updated and tested to ensure their effectiveness against emerging security threats. The big challenge of having effective ML based IDSs is the lack of attack data.

B. GENAI IN UAV SECURITY

UAVs are increasingly targeted by adversarial attacks, including spoofing, jamming, and data injection. Various GAN-based methods have been proposed to detect and mitigate these attacks and to enhance IDS performance. GUIDE [29] and G-IDS [30] systems utilize GANs to generate synthetic data, addressing the challenges of imbalanced datasets and improving IDS robustness. GAN-based models like SeqGAN and LeakGAN have demonstrated significant improvements in attack detection, which increase accuracy up to 37% for known attacks and 30% for unknown attacks.

While GANs have been used to enhance IDS, they are also leveraged to generate adversarial attacks. IDSGAN [31] is a framework designed to create malicious traffic that deceives IDS systems. By dynamically adapting to IDS models, IDSGAN may effectively bypass detection and challenge system robustness. Similarly, GIDS employs GANs to create synthetic data for in-vehicle network IDS, achieving high detection accuracy for known and unknown attacks [32]. Generative models are employed to address the scarcity of security-related data for UAVs, such as the use of Conditional Tabular GANs (CTGAN), Variational Autoencoders (VAE), and Gaussian Copulas to generate realistic jamming attack scenarios. These models enable the training of IDS on diverse and enriched datasets by improving detection capabilities [33]. Moreover, GANs are combined with active learning to dynamically adapt IDS models, improving their effectiveness against evolving threats [32], [33].

Despite these advancements, the grand challenge, the lack of enough attack data, remains. Generative models often struggle with time-series data augmentation, as highlighted in MUVIDS and GUIDE implementations [28], [29]. While GAN-based IDS may detect novel attacks, they require extensive training data and computational resources, which limit their scalability for real-time applications [31], [34].

Recent advances in generative modeling have significantly influenced the field of cybersecurity, particularly in the development of more robust Intrusion Detection Systems (IDS). Diffusion-based models have shown notable performance in generating realistic, high-dimensional network traffic data. For instance, Wang et al. [35] proposed a Denoising Diffusion Probabilistic Model (DDPM)-based intrusion detection system specifically for UAV networks, which demonstrated superior accuracy and robustness compared to traditional methods such as VAE and PCA. In addition, Yang et al. [36] introduced the Diff-IDS framework, a lightweight intrusion detection system that leverages a diffusion model with feature enhancement to handle imbalanced traffic data, outperforming state-of-the-art approaches on benchmarks such as CICIDS2017 and NSL-KDD. Although the study incorporates traditional augmentation techniques (e.g., image flipping), the diffusion model itself is used exclusively in a discriminative role, clearly separating generative and discriminative functions within the framework. A comparative study by Ammara et al. [37] found that CTGAN and CopulaGAN models yielded the highest fidelity and utility scores for generating network traffic data, effectively supporting IDS training. Chalé and Bastian [38] further confirmed the utility of synthetic cyber data generated via GANs and VAEs for training ML-based NIDS, demonstrating that hybrid datasets (real + synthetic) maintained detection accuracy while addressing data scarcity issues.

These contributions validate the strategic use of both Conditional GANs and DDPMs in our proposed framework and substantiate their potential to replicate complex adversarial behaviors while preserving data realism and security relevance.

While previous studies have effectively demonstrated the utility of GAN-based methods such as GUIDE [29], G-IDS [30], SeqGAN, and LeakGAN for enhancing IDS through synthetic data generation and adversarial training, our research extends beyond these approaches in several key ways. Unlike prior work that primarily focuses on a single generative model or application scenario, we systematically evaluate and compare multiple generative models—VAE, Gaussian Copula, CTGAN, and DDPM—to generate synthetic jamming and spoofing attacks specific to UAV systems. Furthermore, in contrast to frameworks like IDSGAN [31] and Diff-IDS [36] that either simulate adversarial traffic or apply a single model to imbalanced datasets, our work introduces a multidimensional evaluation framework that integrates statistical similarity measures (e.g., KL Divergence, Wasserstein Distance, KS Test) with IDS classification metrics to assess the fidelity and impact of synthetic data comprehensively. Table 1 provides a structured comparison between our proposed framework and three

notable GenAI-based IDS approaches: GUIDE, IDSGAN, and Diff-IDS. Each of these works addresses synthetic data generation for intrusion detection, but with different emphases and limitations.

Our inclusion of diffusion models (DDPM) for synthetic attack generation represents a novel contribution in UAV IDS research, building on recent advances but tailoring them specifically to the temporal and spectral complexities of UAV telemetry data. It also differs from previous studies in that it is explicitly applied to both jamming and spoofing attacks in the context of UAV systems, addressing a broader range of threat modalities that are often treated separately or overlooked in earlier research.

Importantly, our multi-train cross-testing strategy evaluates IDS generalization across different combinations of real and synthetic training sets, revealing the strengths and weaknesses of synthetic data under various deployment conditions. This novel integration of generative diversity, security-oriented data augmentation, and systematic cross-evaluation distinguishes our work from earlier GAN-based IDS research.

III. GENAI-BASED JAMMING AND SPOOFING ATTACKS

Generative AI has emerged as a powerful tool for generating synthetic data. In this research, we propose a framework to generate synthetic attack data for UAVs. The framework consists of two parts. Synthetic attack data is generated with generative models in the first part with models that are VAE, Gaussian Copula, CTGAN and DDPM. In the second part, an intrusion detection algorithm is applied to datasets generated in the first part. XGBoost classification is used to distinguish attack data and benign data. The proposed framework is shown in Figure 1.

A. GenAI MODELS

To generate robust and diverse synthetic data for jamming and spoofing attacks, we employed four complementary generative models: VAE, Gaussian Copula, CTGAN, and DDPM. Each model contributes uniquely to capturing statistical structures, temporal behaviors, and multimodal feature dependencies in UAV attack scenarios.

VAE provides a probabilistic framework for learning smooth latent representations, supporting generalization and variability in continuous UAV telemetry data (e.g., velocity, altitude, signal strength).

Gaussian Copula models inter-variable dependencies without strict marginal distribution assumptions. It is efficient and interpretable, making it well-suited for simulating multivariate correlations in UAV data (e.g., GPS and sensor signals) [39].

CTGAN is tailored for tabular datasets with mixed data types and class imbalance. It uses conditional vector encoding and mode-specific normalization to generate realistic and balanced synthetic samples [40].

DDPM iteratively adds and removes noise to model complex, high-dimensional distributions. It effectively captures fine-grained variations in data and is robust against mode collapse [35], [41], [42].

By combining these models, we construct a multi-perspective generative framework that addresses statistical (Gaussian Copula), structural (VAE, CTGAN), and high-fidelity adversarial (DDPM) dimensions of UAV attack modeling.

1) VARIATIONAL AUTOENCODERS

VAE consists of an encoder that compresses the input data x into a latent representation z, defined by parameters (μ) and (σ), which specify a Gaussian distribution. The latent vector z is sampled from this distribution and passed to the decoder, which reconstructs the original data by minimizing reconstruction loss. This enables efficient learning of compact feature representations and supports the generation of diverse synthetic samples.

2) GAUSSIAN COPULA

The Gaussian copula models dependencies between variables without assuming fixed marginal distributions. Initially, metadata and features are integrated using a conditional probability approach to produce an extended dataset. This dataset is then transformed using a Gaussian copula, enabling the generation of synthetic data that preserves correlation structures across variables [39].

3) CONDITIONAL TABULAR GAN

CTGAN is designed to generate synthetic tabular data with mixed feature types. It applies conditioning on discrete columns using logarithmic frequency sampling, enabling it to capture rare feature combinations. The generator receives a noise vector z and conditioning input to produce a synthetic row, which is evaluated by a discriminator. Both networks are fully connected, with the generator using batch normalization and ReLU activation, while the discriminator employs leaky ReLU and dropout. CTGAN enhances TableGAN by incorporating modified loss functions and normalization methods to address imbalanced and non-Gaussian continuous data [43].

4) DENOISING DIFFUSION PROBABILISTIC MODEL

DDPM learns data distributions by gradually corrupting samples with Gaussian noise and then reconstructing them through iterative denoising. The forward process adds noise over K steps, transforming the input X_0 into Gaussian noise X_K. The reverse process reconstructs the original data through a Markov chain of denoising steps. This approach supports high-quality sample generation and is particularly suitable for capturing complex or edge-case attack patterns.

5) ALGORITHMS OF GenAI MODELS

Algorithm 1 outlines the proposed GenAI model for generating synthetic jamming and spoofing attacks in UAVs. The algorithm consists of three main steps. In the first

TABLE 1. Comparison of our framework with existing GenAI-Based IDS approaches.

Feature	GUIDE [29]	IDSGAN [31]	Diff-IDS [36]	Our Framework
Generative Models Used	GAN (single model)	GAN (for adversarial sample generation)	Diffusion (DDPM – discriminative role only)	VAE, CTGAN, Gaussian Copula, DDPM
Application Domain	Time-series emulation	Adversarial traffic generation to bypass IDS	CICIDS2017 network traffic anomaly detection	UAV-specific IDS with jamming/spoofing focus
Attack Types	Flooding and unknown attacks (UAV-specific)	Adversarial samples to deceive IDS	Imbalanced network traffic (not UAV-specific)	Jamming and spoofing (UAV-specific)
Evaluation Strategy	Imbalance mitigation, TPR/FPR metrics	Bypass rate, adversarial success	Benchmark evaluation (accuracy, F1)	Multimodel cross-testing, statistical similarity + IDS metrics
Real+Synthetic Data Fusion	No explicit fusion	No	Not applicable (discriminative use only)	Yes – analyzed in various training-test configurations
Novelty in Modeling	Time-aware GAN for sequence data	IDS-evading sample generation via GAN	Lightweight diffusion model with feature enhancement	First UAV-focused DDPM generator, multi-modal threat generation, statistical fidelity assessment
Generalization Evaluation	Limited to imbalance settings	Focus on adversarial evasion	Focus on lightweight inference	Multi-train cross-testing with real/synthetic splits

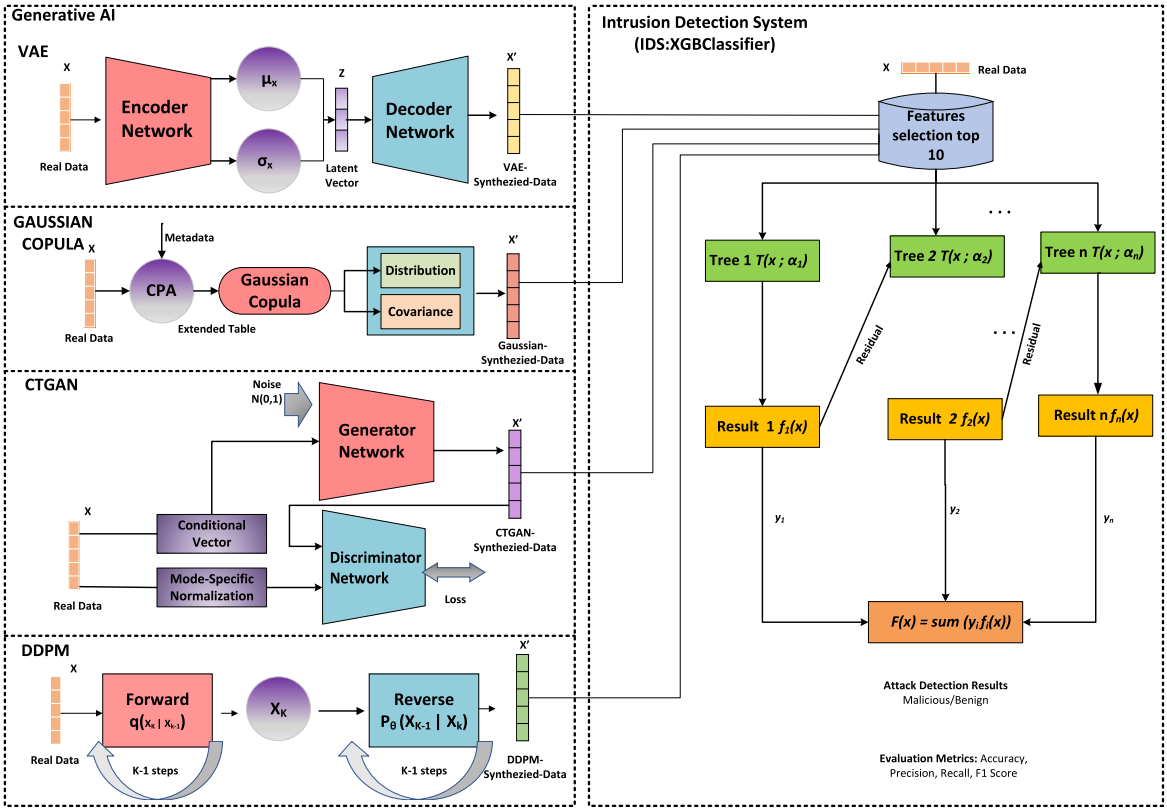


FIGURE 1. Framework for Generating Synthetic Attack Data Using GenAI Models.

step, training a VAE model to generate synthetic attacks is accomplished. Then, we train a Gaussian Copula model to generate synthetic attacks. Finally, we train a CTGAN model to generate synthetic attack, where x_i represents the i-th feature and N is the total number of features and outputs the final generated synthetic attack data for each model.

In step 1, the Variational Autoencoder (VAE) uses the Kullback–Leibler (KL) divergence metric [44], which measures the difference between the learned latent distribution and a target Gaussian distribution. This encourages the latent space to align with the standard Gaussian prior $\mathcal{N}(0, 1)$, promoting regularity and smoothness. Specifically, the KL divergence term quantifies how much the encoder's learned distribution $q(z|x)$ deviates from the prior, as shown in Eq. 1.

Unlike standard autoencoders that map the input x to a single deterministic latent vector, VAEs instead map x to two separate vectors representing the mean (μ) and standard deviation (σ) of a Gaussian distribution. This probabilistic encoding ensures a smooth and continuous latent space, enabling better generalization and generative capabilities [45].

$$L_{\text{KL}} = D_{\text{KL}}(q(z|x) \,\|\, \mathcal{N}(0, 1)) \tag{1}$$

$$L_{\text{total}} = L_{\text{KL}} + \|x - \tilde{x}\|^2 \tag{2}$$

Algorithm 1 Generative Artificial Intelligence for Generated Synthetic Attacks in UAVs

1. **Input:** Training data $x_1, x_2, \ldots, x_n \in X$
2. **Output:** VAE-synthezied-data, GaussianCopula-synthezied-data, CTGAN-synthezied-data, DDPM-synthezied-data
3. **Step 1:Training a VAE model to generating synthetic attacks**
4. Initialize decoder parameters ϕ, encoder parameters θ
repeat
 for $k = 1$ to N **do**
  5. Draw samples S from $\epsilon \sim \mathcal{N}(0, 1)$
  6. Compute latent variable: $z_{(k,s)} = h_\phi(\epsilon^{(k)}, x^{(k)})$
 end for
 7. Compute reconstruction loss: $E = \|x - \tilde{x}\|^2$
 8. Compute KL divergence:
$L_{\mathrm{KL}} = D_{\mathrm{KL}}(q(z|x) \,\|\, \mathcal{N}(0, 1))$
 9. Combine losses: $L_{\text{total}} = L_{\mathrm{KL}} + E$
 10. Update parameters ϕ, θ using Stochastic Gradient Descent
until Parameters ϕ, θ converge
11. **Step 2: Training a Gaussian Copula model to generate synthetic attacks**
for each feature $x_i \in X$ **do**
 12. Estimate the marginal distribution of each feature.
 13. Transform each feature into a standard normal distribution using CDF.
 14. Compute the covariance matrix of the transformed features.
 15. Generate synthetic samples from a multivariate normal distribution using the learned covariance matrix.
 16. Apply the inverse CDF transformation of each feature to map the synthetic data back to the original space.
end for
17. Combine synthetic features.
18. **Step 3: Training a CTGAN model to generate synthetic attacks**
for each feature $x_i \in X$ **do**
 19. Sample noise vector from a standard normal distribution.
 20. Sample conditional vector representing feature distributions.
 21. Generate synthetic samples using the generator.
 22. Discriminate real and synthetic samples using the discriminator.
 23. Compute generator loss as Wasserstein Loss:
$\mathcal{L}(D, G) = \mathbb{E}_{x\sim p_r}\big[D(x)\big] - \mathbb{E}_{z\sim p_g}\big[D(G(z))\big]$
end for

Variational Autoencoder (VAE) loss function ensures the model to learn meaningful latent representations while accurately reconstructing input data. The 1 equation, calculates the Kullback-Leibler (KL) divergence, which measures how much the learned latent distribution deviates from the desired prior distribution, typically a standard normal distribution $\mathcal{N}(0, 1)$. This regularization term enforces structure and continuity in the latent space, enabling smooth interpolation and meaningful sampling. Equation 2 combines the regularization term with the reconstruction loss that is represented as the squared difference between the input x and its reconstruction $\tilde{x}$. This combined objective ensures that the VAE not only produces reconstructions that are faithful to the original input but also learns a well-regularized and structured latent representation. The KL term shapes the latent space, while the reconstruction term preserves input fidelity. Together, these losses guide VAE during training, balancing accurate data reconstruction and a robust latent representation [46].

The training of a Gaussian Copula model to generate synthetic attacks is described in step 2. Each feature is transformed into a uniform distribution through the application of cumulative distribution functions (CDFs), and is then further standardized to a normal distribution through the use of the inverse of the standard normal CDF. A correlation matrix is then derived from the transformed variables in order to capture the relationships between them. The matrix is then employed to generate new samples that are drawn from a multivariate normal distribution. Subsequently, the samples are mapped back to the original variable scales through inverse transformations. Initially, they are smoothed with the normal CDF, and then reverted to the original distributions using the inverse CDFs of the fitted models [47], [48].

The training of a Gaussian Copula model to generate synthetic attacks is described in step 3.

The CTGAN loss function was optimized using the Adam algorithm. For CTGAN, the training process leverages the Wasserstein loss function, which provides a stable and effective objective for generative adversarial learning in the equation 3. In this equation, D represents the discriminator, G denotes the generator, z is a sample from the generator's input noise distribution p_g, and x is a sample from the real data distribution p_r.

$$\mathcal{L}(D, G) = \mathbb{E}_{x\sim p_r}\big[D(x)\big] - \mathbb{E}_{z\sim p_g}\big[D(G(z))\big] \tag{3}$$

This approach offers several advantages over the standard GAN loss: it provides more informative and smooth gradients, reduces training instability, and mitigates issues like mode collapse. Most importantly, it enables the generator to iteratively improve in producing synthetic data that closely approximates the real distribution in both marginal and joint feature spaces.

By minimizing this divergence between real and generated distributions, the Wasserstein loss ensures that the synthetic data retains the structural and statistical properties of the real dataset.

Algorithm 2 outlines the proposed DDPM Model for generating synthetic attacks in UAVs. In our proposed solution algorithm 2 runs after algorithm 1 therefore we continue with step 4 in the second algorithm. The training of a DDPM model to generate synthetic attacks is explained

Algorithm 2 DDPM for Generated Synthetic Attacks

1. **Step 4: Training a DDPM model to generated synthetic attacks:**
for each feature $x_i \in X$ **do**
2. Add Gaussian noise at each step.
3. Train a neural network to predict the noise.
4. Minimize the loss function.
$L_{\text{simple}} = \mathbb{E}_{k,x_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_k, k)\|^2\right]$
5. Sample
6. Iteratively denoise using the reverse process
7. Denormalize the generated data to match the original data distribution
end for
8. **Step 5: Output:**
9. **Return** VAE-synthezied-data, GaussianCopula-synthezied-data, CTGAN-synthezied-data, DDPM-synthezied-data

in step 4. Finally, outputs for each model are recorded in last generated synthetic attack data.

The loss function (in equation 4) L_{simple} is a fundamental component of DDPM models, designed to optimize the model's ability to reconstruct data by accurately predicting the noise introduced during the forward diffusion process. It is defined as the expected mean-squared error (MSE) between the true noise ϵ and the predicted noise $\epsilon_\theta(x_k, k)$, where the prediction is performed by a neural network parameterized by θ [41], [49].

$$L_{\text{simple}} = \mathbb{E}_{k,x_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_k, k)\|^2\right] \tag{4}$$

Let x_k denote the input at time step k, and let $\epsilon \sim \mathcal{N}(0, 1)$ represent the Gaussian noise sampled from a standard normal distribution. The expectation $\mathbb{E}_{k,x_0,\epsilon}$ is computed over the timestamps k, real data samples x_0, and Gaussian noise ϵ. By minimizing this loss, the model learns to predict the noise for any noisy input x_k at a given timestep k. This is crucial for the reverse diffusion process, where the model progressively removes noise to reconstruct the original data.

B. INTRUSION DETECTION SYSTEMS

We tested generated synthetic attack data by using intrusion detection systems for UAVs. We use XGBoost algorithm for the intrusion detection system. XGBoost, a highly scalable tree boosting system, was introduced by Chen and Guestrin [50] as an extended version of the Gradient Boosting Decision Tree (GBDT) algorithm [51].

To reduce the dimensionality of the model while preserving informative features, we used the Gini-based gain importance, which measures the average gain across all splits where a feature is used. For each feature x_i, the gain is calculated as in 5.

$$\text{Gain_Based_Importance}(x_i) = \sum_{t \in T(x_i)} \frac{1}{|T(x_i)|} \Delta\mathcal{L}_t(x_i) \tag{5}$$

where $T(x_i)$ is the set of trees where x_i is used, and $\Delta\mathcal{L}_t(x_i)$ is the reduction in the loss function (e.g., logistic/log-loss or squared error) when splitting on feature x_i at node t. Based on the computed gain scores, features were ranked in descending order of importance. The top ten most important features were selected and used to train the final IDS model, ensuring a balance between performance and computational efficiency. The pseudocode for the feature ranking and selection process is provided in Algorithm 3. According to the ranked list of features, the top ten features were selected for the training of the IDS model.

Algorithm 3 Feature Importance Extraction via XGBoost

Input: Raw feature set $\mathcal{X} = \{x_1, x_2, \ldots, x_n\}$ from training data
Output: Top-k features $\mathcal{X}_{\text{top}}$
1. Compute gain-based importance score $\text{Gain}(x_i)$ for each feature x_i
2. Sort all features in descending order of gain: $\text{Sort}(\mathcal{X}) \rightarrow \mathcal{X}'$
3. Select top-10 features: $\mathcal{X}_{\text{top}} \leftarrow \mathcal{X}'[1:10]$
4. Return $\mathcal{X}_{\text{top}}$

XGBoost operates under the principle of gradient boosting, which consists of three primary stages. The initial step is to identify a differentiable loss function that is appropriate for the problem at this stage of the process. One advantage of gradient boosting is its flexibility, in that new algorithms do not need to be derived for different loss functions. Instead, an appropriate loss function can be selected and integrated into the framework. Secondly, a weak learner is utilized to make predictions, with decision trees being the optimal choice. In particular, regression trees are utilized as weak learners, as they generate real-valued outputs for splits, thereby allowing their outputs to be aggregated. This additive property enables the model to iteratively enhance the accuracy of residuals. The trees are constructed in a greedy manner, frequently with constraints that guarantee they remain weak learners while maintaining computational efficiency. Thirdly, an additive model is constructed by sequentially combining the outputs of the weak learners in order to minimize the loss function. New trees are added to the sequence one at a time, with each tree's output refining the predictions of the previous ones. This process continues until the loss function is sufficiently optimized. XGBoost builds upon gradient boosting but introduces a significant enhancement. In contrast to traditional gradient boosting, where weak learners are added sequentially, XGBoost employs a multi-threaded approach.

A framework for evaluating attack detection using XGBoost with both real and synthesized datasets is shown in Figure 1. From each dataset, the top 10 most important features are selected for training purposes. The XGBoost is then used to model the data. Subsequently, XGBoost, a gradient boosting algorithm, is employed for modeling the

data. The algorithm employs a sequence of decision trees, wherein each tree refines the residual errors of the previous one. The trees are then combined into an additive model, whereby the final prediction is the weighted sum of the outputs of all trees. The framework is designed to predict whether an input sample is ''malicious'' or ''benign.'' Its performance is evaluated using a range of metrics, including accuracy, precision, recall and F1 score. This configuration enables a comparison of the quality of synthetic data to support effective attack detection relative to the real dataset.

Algorithm 4 IDS for Synthetic and Real Attacks on UAVs

Input: Training top 10 important features $x_1, x_2, \ldots, x_i \in X$ extracted from synthetic and real datasets C: Number of trees, P: Set of other parameters of XGBoost classifier
Output: M: Trained XGBoost model
1. Set initial values for the parameters of the XGBoost classifier
2. **for** each instance f_i in F **do**
3. $x_i \leftarrow$ SortFeatures(f_i) Sort features based on feature importance score $G(x_i)$
4. $p_1 \leftarrow \arg\min_{s \in S} \mathcal{L}_{\text{gain}}(s, f_i)$ Select split point s with minimum gain-based loss
5. Optimize objective target function to compute the tree depth and choose the best split point
6. $M \leftarrow$ Optimizing(Choosing descriptor-point(p_i))
7. Tree-leaves $\leftarrow$ Prediction-score(M, Tree-leaves)
8. Bottom-up-Prune-negative-nodes (M, Tree-leaves)
9. Repeat
10. Repeat steps 3 and 4 until Length(M) = Max-tree-depth
11. Repeat steps 3 to 10 until cumulative training includes all trees in C
12. Until
13. Return M
14. **Step 4: Evaluation:**
15. Evaluate final M model using test features.

A method for training an intrusion detection system based on the XGBoost classifier, specifically designed to address synthetic and real attacks on UAVs is described in Algorithm 4. It takes the top 10 important features as inputs that are extracted from both synthetic and real datasets, along with hyperparameters such as the number of trees and other classifier settings. Initially, the parameters of XGBoost classifier are set. For each feature in the dataset, the algorithm sorts the feature values, selects the optimal split point using a lowest-gain-first approach, and optimizes the objective target function to determine the tree depth and split point. Subsequently, the tree structure is updated by optimizing and pruning unnecessary nodes based on their contribution to the prediction task. This process iterates until the maximum tree depth is reached. The cumulative training is repeated for all trees in the dataset, refining the model. Finally, the trained model is returned and evaluated using test data, completing the IDS training process. This approach ensures an efficient and optimized classifier that effectively detects synthetic and real attacks.

IV. ATTACKS AND SCENARIOS

UAVs use a combination of communication networks, sensors, electronic receivers and transmitters, and software systems. UAVs collect and process data, share data over communication networks, control their flight, and perform specific tasks. These systems generate logs that contain substantial volumes of sensitive information, including location coordinates, video streams, sensor readings, and control commands. Attackers can exploit vulnerabilities in these systems to gain unauthorized access, manipulate data, disrupt communications, take control of systems, or cause physical damage.

Dataset contains the UAV's log records during the attack and the flights where no attack occurred. In the scenario implemented by Whelan et al., [52] the UAV must first perform a flight where no attack occurred. Benign flight can be performed in a controlled environment or in a non-hostile area such as a military base. After this flight is completed, it is recorded as benign flight logs. Their scenario used the Global Positioning System (GPS) for navigation of Uncrewed Aerial Vehicles (UAVs) and the MAVLink protocol for communication with the Ground Control Station (GCS) [12]. These systems are vulnerable to certain threats, especially jamming and GPS spoofing. Spoofing disrupts communication or GPS signals, while GPS spoofing involves broadcasting false GPS signals to deceive the UAV's navigation system. Figure 3 illustrates a typical scenario involving a Ground Control Station (GCS) and an Uncrewed Aerial Vehicle (UAV) communicating via the MAVLink protocol. The scenario involves a UAV that communicates with a GCS using the MAVLink protocol, a widely used messaging protocol for UAVs.

MAVLink is a lightweight protocol designed for communication links with limited bandwidth and systems with constrained resources, which are used to carry telemetry and command and control data for numerous small uncrewed aerial vehicles [53]. MAVLink protocol defines the mechanism on the structure of messages and how to serialize them at the application layer. These messages are then forwarded to lower layers, such as transport layer and physical layer, to be transmitted to the network. The advantage of MAVLink protocol is that it supports different types of transport layers and mediums thanks to its lightweight structure. It can be transmitted through WiFi, Ethernet, or serial telemetry low bandwidth channels.

Figure 2 describes the structure and meaning of each component in a MAVLink frame, which is a communication protocol widely used in UAVs. The first byte is the ''Packet Start Sign'' (STX), which has a fixed value of 0xFE and signals the beginning of a new packet. The ''Payload Length'' (LEN) indicates the size of the following payload, ranging from 0 to 255 bytes. The ''Packet Sequence'' (SEQ) is used for tracking the sequence of packets to

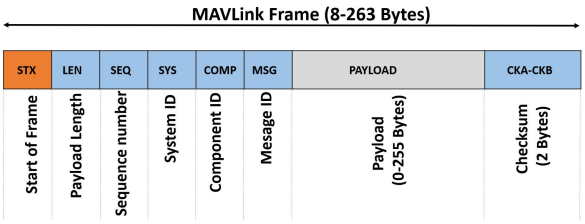


FIGURE 2. MAVLink Protocol Data Frame.

help detect any packet loss. The "System ID" (SYS) and "Component ID" (COMP) identify the sending system and its specific component, allowing for differentiation between multiple platforms and components on the same network. The "Message ID" (MSG) denotes the type of message, which determines the payload's meaning and how to interpret it. The "Data" (Payload), ranging from 0 to 255 bytes, starts depends on the message ID. Lastly, the checksum ensures data integrity by computing a hash of the previous bytes, incorporating an extra parameter to enhance error-checking. This structured format helps MAVLink achieve reliable and organized communication in real-time applications [54].

MAVLink protocol has no privacy or authentication mechanism. Therefore, it does not provide any security and can be hacked quite easily. GCS communicates with drones over an unauthenticated and unencrypted channel. Since MAVLink message streams are sent without encryption, anyone with a suitable transmitter can communicate with the drone, intercept communication, eavesdrop, and inject commands. Thus, any attack can be launched easily. In [55] discusses the vulnerabilities of MAVLink protocol and proposes a security-integrated mechanism for MAVLink that ensures the protection of MAVLink messages exchanged between UAVs and GCSs by utilizing the use of encryption algorithms.

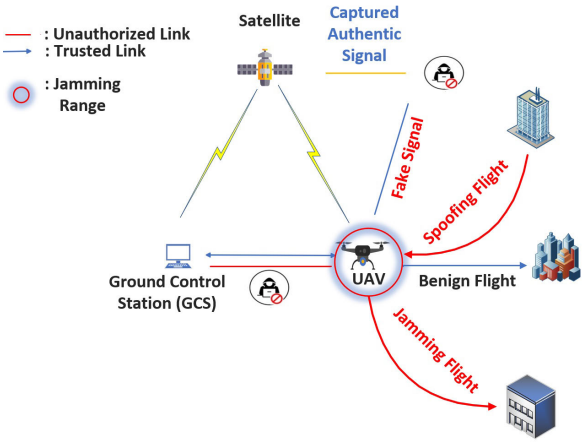


FIGURE 3. An attack scenario.

A Holybro S500 UAV equipped with a Pixhawk 4 flight controller was used to explore UAV intrusion detection under GPS spoofing and jamming attacks in [12]. A HackRF software-defined radio (SDR) performed the attacks, while a Keysight EXG signal generator provided controlled GPS signals simulating a location. The UAV first performed a harmless flight to collect basic data. During the GPS spoofing attack, the GPS-SDR-SIM generated false GPS data streams, causing the UAV to lose its balance and crash. Similarly, in the GPS jamming attack, white Gaussian noise corrupted the UAV's GPS, again leading to a crash. Observations from these harmless and harmless flights were recorded for further analysis. The log records of all flights were shared openly, shedding light on our work. We generated synthetic data for both benign and attack classes using generative AI, based on the data collected in the attack scenario described above. We then compared the statistical properties and intrusion detection performance of our generated dataset with the real dataset from [12].

V. EXPERIMENTAL SETUP

In our research, PyTorch v1.13.0 framework was used for training and evaluating models.

A. DATASETS AND FEATURES

UAV attack dataset in [52] provides a comprehensive repository of real communication data between UAVs and ground control stations, providing a valuable basis for simulating various attack scenarios, including GPS spoofing and jamming attacks. Moreover, the dataset includes both simulated and real flight records, and data from benign flights and flights under attack. The dataset typically includes telemetry data, commands, and status messages exchanged between a UAV and a ground control station. Data covers both benign and malicious examples derived from real flight records, reaching a total of 6,446 rows and 84 columns for jamming attacks and 3,623 rows and 84 columns for GPS spoofing attacks. Table 2 summarizes the number of benign and malicious examples for each dataset, grouped by spoofing and jamming attack types. This table also presents a comparison of benign and malicious examples between real and generated synthetic datasets.

TABLE 2. Comparison of benign vs. Malicious samples across real and synthetic datasets.

Attack Type	Dataset	Benign	Malicious
Spoofing	Real Data	3124	498
	VAE Generated Data	3456	2804
	Gaussian Generated Data	2984	3236
	CTGAN Generated Data	3340	2860
	DDPM Generated Data	1761	1861
Jamming	Real Data	2561	596
	VAE Generated Data	2815	2485
	Gaussian Generated Data	2700	2600
	CTGAN Generated Data	2990	2310
	DDPM Generated Data	2890	2410

Figure 4 presents the distribution of benign and malicious samples in the jamming attack dataset, where 22.7% of the data is malicious and 77.3% is benign. Similarly, Figure 5 shows the distribution for the spoofing attack dataset, in which 13.7% of the data is malicious and 86.3% is benign. Both figures include comparisons of synthetic and real data,

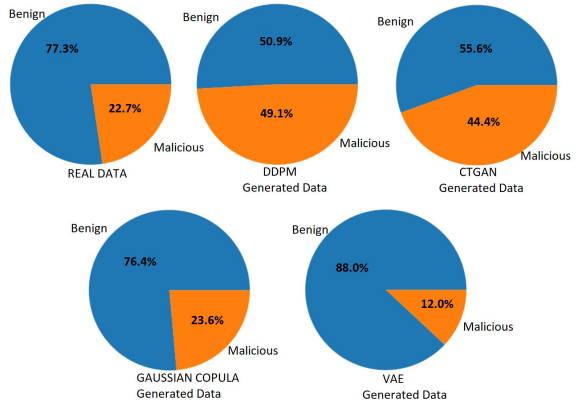


FIGURE 4. Distribution of benign and malicious samples in jamming attack datasets.

highlighting the malicious-to-benign sample ratios for each attack scenario.

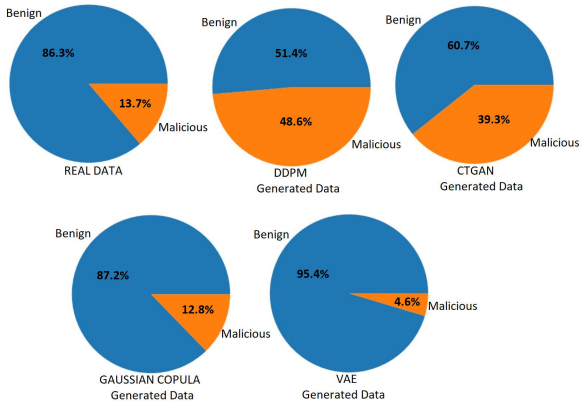


FIGURE 5. Distribution of benign and malicious samples in spoofing attack datasets.

We used XGBOOST classifiers to extract important features from the real dataset. The provided feature importance data explains how certain features play important roles in identifying jamming and spoofing attacks in GPS systems.

These two types of attacks have different impacts on GPS signal quality and navigation data, and the extracted features provide insights into the detection of these attacks and the generation of attacks with similar features to generative models. For example, jamming attack focuses on degrading GPS signal quality, which leads to a noise, a reduced satellite utilization, and a poor positional accuracy. Spoofing attack actively manipulates GPS data to mislead users, which results in impacts on location, speed, and orientation.

Important features of GPS jamming attacks identified by the XGBOOST classifier are given in Figure 6. In the figure, *eph_x* represents the horizontal position accuracy of the GPS signal. *epv* indicates the vertical position accuracy. These features achieve the highest feature importance scores, highlighting their strong predictive power in identifying jamming. They measure the deviations in GPS position accuracy that jamming significantly increases.

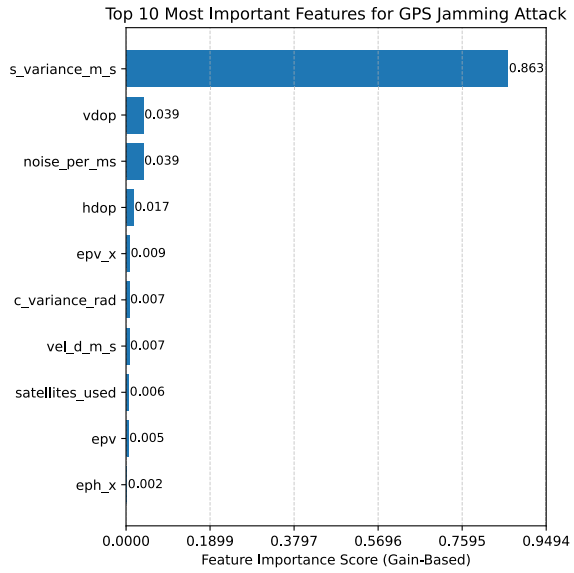


FIGURE 6. Important features for jamming attacks.

Feature *satellites_used* denotes the number of satellites utilized in the GPS positioning system. The feature importance score of this feature indicates its importance in detecting a loss of usable satellites, which is sign of jamming. A reduction in the number of active satellites has a significant impact on the reliability of the GPS system. *vel_d_m_s* feature measures vertical velocity in meters per second. The high feature importance value of this feature suggests that jamming disrupts vertical velocity calculations, creating noisy data.

Feature *c_variance_rad* represents the variance of the path over the ground. The predictive power of this feature reflects the effect of obstructions on the stability of the computed trajectory. *hdop* and *vdop* features indicate the quality of the horizontal and vertical position solution. With significant feature importance values, these features show how jamming undermines the accuracy of GPS positioning by affecting satellite geometry. *noise_per_ms* feature measures the noise in the GPS signal. The high feature importance value of this feature indicates its effectiveness in capturing the increased signal noise associated with jamming. *s_variance_m_s* feature is the variance of the calculated speed. A high feature importance score for this feature indicates that it is useful for identifying irregular speed fluctuations caused by jamming.

In summary, if we consider effects of jamming features, jamming attacks degrade GPS signal quality by creating noise and reducing the number of effective satellites. Features such as *hdop*, *vdop*, and *noise_per_ms* effectively capture this degradation. Positional and velocity inaccuracies (*epv_x*, *epv*, *vel_d_m_s*) provide additional evidence of jamming attack.

Important features of GPS spoofing attacks identified by the XGBOOST classifier are given in Figure 7. In the figure, *eph* represents the horizontal position error similar to *eph_x* in jamming. This feature has a high feature importance

score indicating its effectiveness in detecting positional inaccuracies.

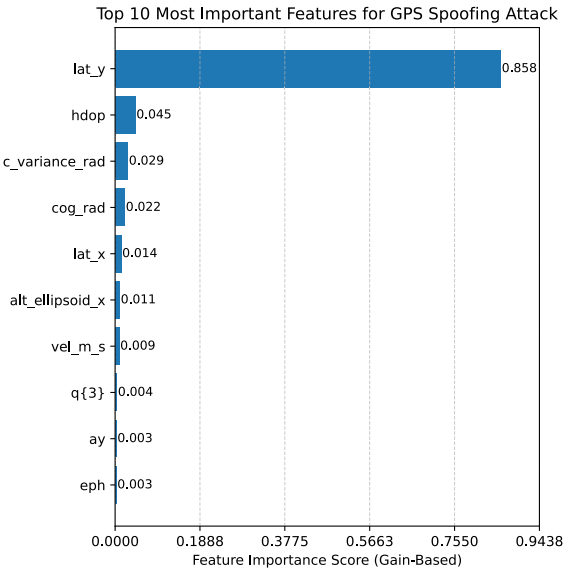


FIGURE 7. Important features for spoofing attacks.

The feature *ay* represents the acceleration along the y-axis. *q*{3} represents the 3D orientation in space. These features achieve high feature importance scores, demonstrating their ability to detect the inconsistencies in motion and orientation caused by spoofed signals. *vel_m_s* feature is the overall velocity in meters per second. The feature importance score here indicates its sensitivity to altered velocity data typical of spoofing attacks.

alt_ellipsoid_x feature represents the GPS altitude relative to a reference ellipsoid. This feature's feature importance score reflects its usefulness in detecting spoofed altitude values that differ from actual measurements. *lat_x* and *lat_y* are latitude coordinates. With remarkable feature importance scores, these features demonstrate their effectiveness in capturing false positional information injected by spoofing. *cog_rad* represents the direction of travel in radians. A high feature importance score for this feature emphasizes its role in detecting spoofing changes to the directional data. *c_variance_rad* feature measures course consistency. This feature captures inconsistency in course calculations caused by spoofing based on the feature importance score. *hdop*, like jamming, measures horizontal accuracy. Unlike jamming, spoofing can artificially reduce this value to create a false sense of precision that the feature successfully identifies.

In summary, when considering the impact of spoofing features on spoofing detection, features related to a position, a motion, and an orientation are the most important. Their high feature importance values confirm their ability to detect the manipulated data injected during spoofing attacks.

Analysis of feature importance scores reveals distinct patterns in the importance of features for detecting jamming and spoofing attacks. Jamming features emphasize the signal degradation and the noise, while spoofing features emphasize the positional and the navigational inconsistencies. This distinction allows us to create targeted detection strategies for these types of GPS attacks. Thus, extracted features are proven to be defining features of the attacks according to the feature importance score. These defining features helped us both in attack detection and in similarity tests of the generated synthetic data similar to original data.

B. GENERATIVE AI MODELS CONFIGURATION

In this section, we discuss the architectural settings, hyperparameters, and baselines used for each generative model.

- VAE, we employ TVAE Synthesizer framework that uses a variational autoencoder [40]. It has different hyperparameters related to neural network like *batch_size*, *compress_dims*, and *decompress_dims*. These settings are specific to the neural network and are intended for optimizing the technical architecture and modeling process. *batch_size* parameter defines the number of data samples processed in each step with a default value of 10. Each model is trained for up to 300 *epochs*. *compress_dims* parameter specifies the size of each hidden layer in the encoder, with a default value of (128, 128). *decompress_dims* determines the size of each hidden layer in the decoder with default values (128, 128). *embedding_dim* parameter sets the embedding dimension size used by both the encoder and decoder, with a default value of 128. Regularization is controlled by *l2scale* parameter, which defaults to 1×10^{-5}. Lastly, *loss_factor* parameter acts as a multiplier for the reconstruction error, with a default value of 2.
- Gaussian Copula, we used Gaussian Copula Synthesizer, synthetic data vault (SDV), framework for the implementation [39]. First, the synthesizer learns the marginal distribution for each individual column, such as a beta distribution with parameters $\alpha = 2$ and $\beta = 5$. It uses this learned distribution to normalize the column values, transforming them into standard normal distributions with a mean of $\mu = 0$ and a standard deviation of $\sigma = 1$. Next, the synthesizer learns the covariance between each pair of normalized columns. These covariances are stored in an $n \times n$ matrix, where n represents the number of columns in the dataset.
- CTGAN, we employ fully-connected networks in both the generator and discriminator to capture all possible correlations among the columns. The generator and discriminator each utilize *two fully-connected hidden layers*. In the generator, batch normalization and *ReLU activation functions* are applied. Following the two hidden layers, the generator produces synthetic row representations using a combination of activation functions. Scalar values are generated with a *tanh activation*, while mode indicators and discrete values are generated using a *Gumbel softmax function*. In the discriminator, each hidden layer employs leaky *ReLU activation functions* and incorporates *dropout* for regularization [56].

- DDPM, models are configured with different number of neurons [256, 512, 1024, 2048, 4096, 8192] and hidden layers [4, 6, 8, 10, 12]. Each model is trained for up to 3000 epochs with a minibatch size of 512. Adam optimizer is employed for training, with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, alongside a cosine learning rate scheduler. The model weights are initialized randomly following the method described in [57].

C. IDS CONFIGURATION

The XGBoost classifier (version 2.1.0) was hyperparameter-tuned via a grid search over the key tree parameters *learning_rate*, *max_depth*, and *min_child_weight*, as shown in Tables 3 and 4. For each attack type (jamming vs. spoofing) and each training-set variant, we selected the combination that maximized 10-fold cross-validation accuracy. All other parameters were left at their XGBoost defaults *booster type* is set to $'gbtree'$, which means the model uses tree-based methods. Since our classifier performs binary classification as malicious/benign, the objective function is $'binary : logistic'$. *verbosity* level is set to 1, thereby ensuring that any relevant warning messages are displayed and that *use_label_encoder* parameter is *enabled*, which applies label encoding to the target variable.

TABLE 3. Optimal hyperparameters for GPS jamming IDS.

Training Set	learning_rate	max_depth	min_child_weight
Real	0.10	3	1
Real + VAE	0.10	7	3
Real + Gaussian Copula	0.10	7	1
Real + DDPM	0.10	7	3
Real + CTGAN	0.10	7	1

TABLE 4. Optimal hyperparameters for GPS spoofing IDS.

Training Set	learning_rate	max_depth	min_child_weight
Real	0.01	3	1
Real + VAE	0.10	7	1
Real + Gaussian Copula	0.10	7	5
Real + DDPM	0.10	7	3
Real + CTGAN	0.10	7	1

This grid-search procedure ensures that each IDS instance is tailored to the characteristics of its training data, improving both detection performance on in-distribution samples and robustness against synthetic attack variants.

For the cross-testing experiments, each dataset variant was randomly split into 70% for training and 30% for testing. Stratified sampling was used to preserve the malicious/benign class ratio in both subsets. The XGBoost classifier was trained on the 70% split using the optimal hyperparameters from the grid search, and all performance metrics were evaluated on the held-out 30%.

D. EVALUATION METRICS

We use the following metrics to evaluate the proposed solution.

- **Kullback–Leibler (KL) Divergence:** KL divergence is a measure of the discrepancy between two probability distributions, quantifying the extent to which the synthetic data generated deviates from the actual data set.

$$D_{KL}(P\|Q) = \sum_{x\in\mathcal{X}} P(x)\cdot \log\frac{P(x)}{Q(x)} \tag{6}$$

- **Wasserstein Distance:** Wasserstein distance quantifies the difference between two probability distributions. In the case of one-dimensional distributions, it is formally defined as follows.

$$W(P,Q) = \int_{x\in\mathbb{R}} \left|P(x) - Q(x)\right| dx \tag{7}$$

- **Kolmogorov-Smirnov Statistic (KS):** KS is a measure used to determine whether two samples originate from the same distribution. It is calculated as the largest absolute difference between the cumulative distribution functions (CDFs) of the two samples, denoted as F_1 and F_2.

$$D = \max_x \left|F_1(x) - F_2(x)\right| \tag{8}$$

D quantifies the disparity between two CDFs. A p-value is then derived from KS statistic to assess the statistical significance of the observed difference.

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{9}$$

The evaluation of classification models often employs a range of metrics, including accuracy, precision, recall, and F1 score. The accuracy metric assesses the overall accuracy of the model by calculating the ratio of correctly predicted examples (true positives and true negatives) to the total number of examples. However, accuracy can be misleading in imbalanced datasets, so it is necessary to evaluate precision and recall as well.

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{10}$$

Precision is defined as the ratio of true positive predictions among all positive predictions. It indicates the accuracy of a model in identifying the positive class.

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

Recall quantifies the proportion of true positive examples that are correctly identified by the model, thereby highlighting its capacity to detect positive cases.

$$\text{F1 Score} = 2\cdot\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} \tag{12}$$

The F1-score is a single metric that combines the precision and recall values through a harmonic mean calculation.

$$\text{TPR} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{13}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP}+\text{TN}} \tag{14}$$

TABLE 5. Similarity of important features between real and synthetic data for GPS jamming attack.

Metric	Important Features	CTGAN	VAE-GAN	Gaussian Copula	DDPM
KL Divergence	*eph_x*	**1.234**	2.345	1.567	1.789
	epv	0.987	1.234	0.876	**0.765**
	satellites_used	**0.567**	2.345	1.234	0.987
	vel_d_m_s	2.345	3.456	2.123	**1.890**
	c_variance_rad	1.456	2.678	1.345	**0.987**
	epv_x	0.890	1.234	0.765	**0.654**
	hdop	0.543	0.678	0.432	**0.321**
	noise_per_ms	**1.234**	2.345	1.678	**1.234**
	vdop	0.987	1.456	0.876	**0.654**
	s_variance_m_s	0.678	1.234	0.543	**0.432**
Wasserstein Distance	*eph_x*	0.234	0.345	0.123	**0.098**
	epv	0.123	0.234	0.098	**0.076**
	satellites_used	0.456	0.678	0.432	**0.321**
	vel_d_m_s	0.567	0.789	0.543	**0.432**
	c_variance_rad	0.234	0.456	0.321	**0.210**
	epv_x	0.098	0.123	0.076	**0.054**
	hdop	0.321	0.432	0.210	**0.098**
	noise_per_ms	0.654	0.789	**0.543**	0.432
	vdop	0.210	0.321	0.098	**0.054**
	s_variance_m_s	0.098	0.123	0.076	**0.054**
KS Statistic	*eph_x*	0.543	0.654	0.432	**0.321**
	epv	0.321	0.543	0.234	**0.123**
	satellites_used	0.789	0.890	0.678	**0.543**
	vel_d_m_s	0.432	0.543	0.321	**0.234**
	c_variance_rad	0.543	0.678	0.432	**0.321**
	epv_x	0.321	0.432	0.234	**0.123**
	hdop	0.654	0.789	0.543	**0.432**
	noise_per_ms	0.543	0.654	0.432	**0.321**
	vdop	0.432	0.543	0.321	**0.234**
	s_variance_m_s	0.321	0.432	0.234	**0.123**

In combination, these metrics provide a comprehensive assessment of model performance, highlighting the various aspects of predictive accuracy and robustness.

VI. EXPERIMENTAL ANALYSIS

In this section, statistical results of synthetic attacks generated with GenAI models are analyzed. IDS performance results are given for both spoofing and jamming attacks with the generated synthetic attack data.

A. GenAI STATISTICAL RESULTS

1) GPS JAMMING ATTACK GENERATION RESULTS

Table 5 contains a detailed comparison of synthetic data generation methods CTGAN, VAE-GAN, Gaussian Copula, and DDPM for features in jamming scenarios. Metrics used for the evaluation include KL Divergence, Wasserstein Distance, and KS Statistic.

VAE struggles to match the real data distribution, showing significantly higher KL Divergence for variables like *vel_d_m_s* and *c_variance_rad*, suggesting its synthetic data is less aligned with the real data. VAE also shows higher Wasserstein Distances across most variables, with notable gaps for *vel_d_m_s* and *c_variance_rad*, confirm its weaker alignment with the real data. VAE consistently performs poorly, with higher KS Statistic scores across all variables, showing significant distributional differences between its synthetic data and the real data.

Gaussian Copula achieves KL Divergence values close to those of CTGAN for the variable *eph_x*. It also yields competitive Wasserstein Distance values, particularly for the variable *noise_er_ms*. In terms of KS Statistic, Gaussian Copula produces consistent results across key features. These findings indicate that Gaussian Copula can replicate certain variable distributions with a high degree of similarity to the real data. Among the evaluated models, Gaussian Copula generates data most similar to the real dataset after DDPM.

CTGAN consistently achieves the lowest KL Divergence for most variables, such as *eph_x*, *satellites_used*, and *noise_per_ms* indicating its strength in closely approximating the distribution of real data.

TABLE 6. Synthetic data comparison for GPS jamming attack.

Feature	Real Data	CTGAN	DDPM	Gaussian Copula	VAE-GAN
eph_x	1.23 ± 0.34	1.25 ± 0.36	1.21 ± 0.33	1.22 ± 0.35	1.27 ± 0.37
epv	0.98 ± 0.12	0.97 ± 0.11	0.96 ± 0.10	0.99 ± 0.13	0.95 ± 0.14
satellites_used	5.12 ± 1.34	5.15 ± 1.35	5.10 ± 1.33	5.08 ± 1.32	5.13 ± 1.36
vel_d_m_s	2.45 ± 0.54	2.50 ± 0.55	2.42 ± 0.52	2.46 ± 0.53	2.48 ± 0.56
c_variance_rad	0.23 ± 0.06	0.24 ± 0.07	0.22 ± 0.05	0.25 ± 0.08	0.21 ± 0.07
epv_x	0.34 ± 0.09	0.33 ± 0.08	0.35 ± 0.10	0.32 ± 0.09	0.36 ± 0.10
hdop	1.12 ± 0.31	1.13 ± 0.32	1.10 ± 0.30	1.15 ± 0.33	1.14 ± 0.34
noise_per_ms	0.45 ± 0.12	0.46 ± 0.13	0.44 ± 0.11	0.47 ± 0.14	0.43 ± 0.12
vdop	0.67 ± 0.18	0.66 ± 0.17	0.68 ± 0.19	0.69 ± 0.20	0.65 ± 0.18
s_variance_m_s	0.78 ± 0.22	0.80 ± 0.23	0.76 ± 0.21	0.79 ± 0.24	0.77 ± 0.22

DDPM demonstrates strong performance with the lowest KL Divergence, Wasserstein Distance and, KS Statistic for all important features. This highlights DDPM's capability to minimize the distributional difference between real and synthetic data. It indicates a closer match to the cumulative distribution of real data.

The evaluation of synthetic data generation methods for jamming scenarios highlights DDPM as the most reliable and consistent performer across all metrics, particularly excelling in KL Divergence and Wasserstein Distance. Its ability to closely replicate real data distributions for critical variables. Gaussian Copula provides competitive results for selected variables, such as *c_variance_rad* and *vel_d_m_s*, which indicates its suitability for these distributions. However, VAE consistently under-performs, with higher KL Divergence and Wasserstein Distance values across most variables, underscoring its limitations in accurately capturing real data characteristics. These findings emphasize the importance of selecting the appropriate synthetic data generation method based on the specific application, with DDPM emerging as the most robust choice, while CTGAN and Gaussian Copula offer valuable alternatives for variable-specific requirements.

CTGAN emerges as the most reliable model, consistently producing synthetic data with mean and standard deviation values closest to the real data across most variables. For instance, variables like *eph_x*, *satellites_used*, and *hdop* show minimal deviations, reflecting CTGAN's robust ability to replicate the central tendency and variability of real data distributions. This makes CTGAN particularly suitable for tasks that demand precise statistical replication.

DDPM performs competitively, especially for variables such as *vel_d_m_s* and *s_variance_m_s*, where it closely approximates the real data in terms of both mean and standard deviation. However, for more complex variables like *c_variance_rad* and *noise_per_ms*, DDPM demonstrates slightly larger deviations, suggesting limitations in modeling distributions with higher variability.

Gaussian Copula shows relatively close mean and standard deviation values to real data for variables such as *epv* and *vdop*. However, its performance is inconsistent, as evident in variables like *epv_x* and *satellites_used*, where notable deviations are observed, suggesting limitations in modeling complex or multimodal distributions.

In VAE generating plausible data for some variables such as *noise_per_ms*, exhibits significant deviations in both mean and standard deviation for variables like *c_variance_rad* and *hdop*. This underperformance highlights limitations in its architecture, making it less effective for accurately replicating statistical properties of real data in jamming scenarios. Table 6 summarizes the mean and standard deviation of the specified variables for real data and synthetic data generated by CTGAN, DDPM, Gaussian Copula, and VAE has been calculated.

In general, Gaussian Copula demonstrates the most consistent and reliable performance, excelling at reproducing real data distributions across most variables. DDPM shows promise for certain features but lacks the overall consistency of Gaussian Copula. CTGAN delivers competitive results, especially for features like *eph_x* and *satellites_used*, but slightly under-performs on others. VAE exhibits strong performance for some variables but requires further optimization to achieve the consistency observed in Gaussian Copula.

The distributions of three selected important features are shown as density plots on the graphs for each model, VAE, Gaussian Copula, CTGAN, DDPM, in the jamming attack. The density plot demonstrates the distribution of the *s_variance_m_s* feature across the datasets comprising the real data, CTGAN, Gaussian Copula, DDPM, and VAE-GAN models in the context of a GPS jamming attack in Figure 8. The real data, depicted by the blue curve, represents the ground truth and exhibits an uni-modal distribution with a peak occurring at approximately 0.8. Among the generative models, Gaussian Copula, green curve, most closely resembles the real data, exhibiting a peak at nearly the same location and a similar spread.

VAE, purple curve, shows a broader distribution, deviating from the compactness of the real data while slightly overestimating the density at higher values. CTGAN, orange curve, demonstrates a reasonable alignment with the real data,

though it shifts the peak slightly to the right and broadens the spread. DDPM, red curve, captures the general shape but underestimates the density at the peak while overestimating it in the tails. This analysis suggests that Gaussian Copula is the most effective model for replicating this feature, while VAE and DDPM require further refinement to improve their fidelity. CTGAN offers a moderate balance between fidelity and diversity.

The density plot demonstrates the distribution of *epv_x* feature across the datasets in GPS jamming attacks in Figure 9. The real data, blue curve, exhibits a sharp, concentrated peak at approximately 4.375. Among the generative models, DDPM, red curve, aligns most closely with the real data, accurately replicating the peak density and shape.

The Gaussian Copula model (green curve) closely approximates the real distribution but exhibits a slight broadening, indicating reduced fidelity. CTGAN (orange curve) shows a noticeable shift in the peak and a wider spread, diverging more substantially from the reference distribution. VAE (purple curve) presents further deviations, with discrepancies in both peak density and overall shape. Among the models evaluated, DDPM most closely matches the target distribution for this feature, followed by Gaussian Copula. CTGAN and VAE show larger deviations, suggesting potential areas for improvement in modeling this variable.

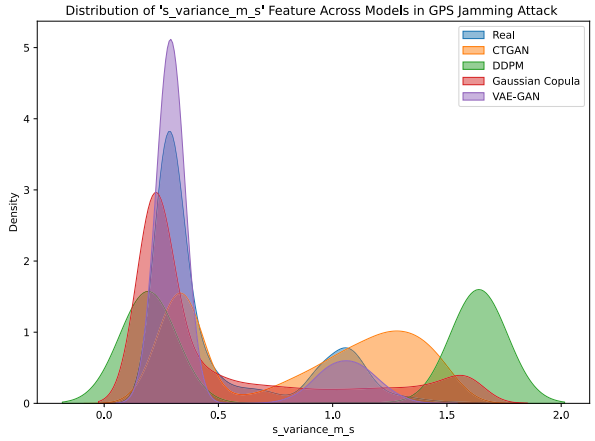


FIGURE 8. Distributions of 's_variance_m_s' feature across models in GPS jamming attacks.

The density plot for *noise_per_ms* feature highlights the distribution modeling challenge, with real data, blue curve, that shows an uni-modal distribution centered around 100 in Figure 10. Among the generative models, Gaussian Copula, green curve, replicates the general shape of the real data but introduces slightly broader variability and subtle secondary peaks, indicating moderate fidelity.

VAE, purple curve, captures the peak but exhibits sharper curves and slightly reduced spread, which might suggest over-fitting and limited diversity. CTGAN, orange curve, produces a broader distribution with a peak shifted slightly to the right, showing a moderate deviation from the real data.

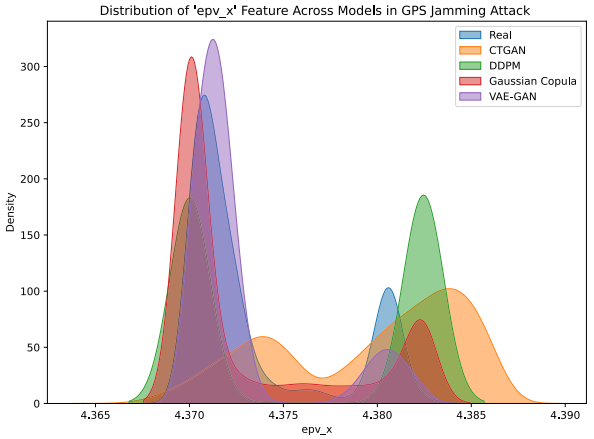


FIGURE 9. Distributions of 'epv_x' feature across models in GPS jamming attacks.

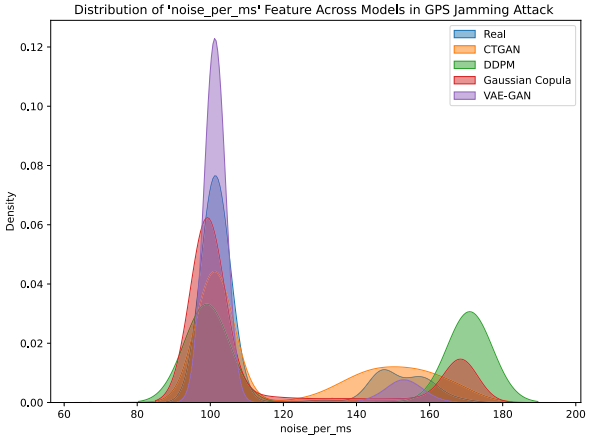


FIGURE 10. Distributions of 'noise_per_ms' feature across models in GPS jamming attacks.

DDPM, red curve, also shows broader variability and slightly under-represents the peak density.

Analyses results suggest that Gaussian Copula provides the best overall balance between fidelity and diversity for this feature, while VAE offers high fidelity but limited diversity. CTGAN and DDPM demonstrate moderate performance but deviate in peak location and spread, which indicates areas for improvement.

2) GPS SPOOFING ATTACK GENERATION RESULTS

VAE struggles with consistently higher KL Divergence values, particularly for $q\{3\}$ and *cog_rad*, which indicates a poor alignment with the real data distributions. VAE underperforms, with higher Wasserstein Distance values for most variables, especially *cog_rad* and *lat_x*. Furthermore, it indicates significant distributional differences. VAE consistently shows higher KS Statistic values, particularly for *cog_rad* and *lat_x*, reflecting significant discrepancies in distribution alignment.

Gaussian Copula KL divergence values are close to DDPM in capturing some features, on the other hand, it falls behind CTGAN in capturing some features (*eph*, *vel_m_s*,*cog_rad*,

TABLE 7. Similarity of features between real and synthetic data for GPS spoofing attacks.

Metric	Important Features	CTGAN	VAE	Gaussian Copula	DDPM
KL Divergence	*eph*	**1.123**	2.456	1.789	1.987
	ay	0.876	1.234	0.765	**0.654**
	q{3}	2.456	3.789	2.123	**1.890**
	vel_m_s	**0.765**	1.234	0.987	0.876
	alt_ellipsoid_x	1.345	2.456	1.234	**0.987**
	lat_x	0.543	0.678	0.432	**0.321**
	cog_rad	1.678	2.345	1.890	**1.234**
	c_variance_rad	**1.234**	2.345	1.567	1.432
	hdop	0.987	1.234	0.876	**0.765**
	lat_y	0.678	1.234	0.543	**0.432**
Wasserstein Distance	*eph*	0.234	0.345	0.123	**0.098**
	ay	0.123	0.234	0.098	**0.076**
	q{3}	0.567	0.789	0.543	**0.432**
	vel_m_s	0.321	0.432	0.210	**0.098**
	alt_ellipsoid_x	0.654	0.789	0.543	**0.432**
	lat_x	0.098	0.123	0.076	**0.054**
	cog_rad	0.210	0.321	0.098	**0.054**
	c_variance_rad	0.098	0.123	0.076	**0.054**
	hdop	0.234	0.345	0.123	**0.098**
	lat_y	0.210	0.321	0.098	**0.054**
KS Statistic	*eph*	0.543	0.654	0.432	**0.321**
	ay	0.321	0.543	0.234	**0.123**
	q{3}	0.789	0.890	0.678	**0.543**
	vel_m_s	0.432	0.543	0.321	**0.234**
	alt_ellipsoid_x	0.543	0.678	0.432	**0.321**
	lat_x	0.321	0.432	0.234	**0.123**
	cog_rad	0.654	0.789	0.543	**0.432**
	c_variance_rad	0.543	0.654	0.432	**0.321**
	hdop	0.432	0.543	0.321	**0.234**
	lat_y	0.321	0.432	0.234	**0.123**

c_variance_rad). Gaussian Copula achieves notable KS Statistic values for key features, demonstrating its capability to accurately replicate certain variable distributions. Additionally, Gaussian Copula is the second most effective model, producing synthetic data closely aligned with real data, following DDPM.

CTGAN exhibits strong performance across most variables, achieving the lowest KL Divergence values for variables such as *eph*, *vel_m_s* and *c_variance_rad*. This demonstrates its capability to approximate real data distributions closely.

DDPM exhibits outstanding performance, achieving the lowest KL Divergence, Wasserstein Distance, and KS Statistic in almost all key features. This underscores DDPM's ability to effectively minimize distributional differences between real and synthetic data, indicating a closer alignment with the cumulative distribution of the real data. Table 7 contains a detailed comparison of synthetic data generation methods, CTGAN, VAE, Gaussian Copula, and DDPM for important features in jamming scenarios.

DDPM emerges as the most reliable synthetic data generation method for spoofing scenarios, consistently outperforming other methods in replicating distributions of real data. DDPM provides strong results for select variables, showcasing its potential for variable-specific applications. VAE requires substantial improvement, as it consistently fails to align its synthetic data with real data distributions. These findings reinforce the importance of selecting the appropriate synthetic data generation method based on the specific variables and metrics critical to the application.

CTGAN demonstrates moderate performance but it does not consistently achieve the closest approximation to real data. While it performs reasonably well for variables like *eph* and *lat_x*, it exhibits noticeable deviations for variables such as *hdop* and *c_variance_rad*. These deviations suggest that

while CTGAN can capture central tendencies, it struggles with higher accuracy in complex variable distributions.

DDPM performs strongly overall with closely matching the real data for variables like *vel_m_s* and *alt_ellipsoid_x*. However, its performance slightly lags for variables like *cog_rad*, where deviations are small but more noticeable compared to Gaussian Copula. DDPM consistently minimizes errors in both mean and variance, making it a strong contender, especially for features with moderate complexity.

Gaussian Copula demonstrates moderate accuracy, particularly excelling in variables like *c_variance_rad* and *q{3}*. However, its higher variability in key metrics such as *lat_x* and *hdop* indicates limitations in its ability to model distributions with higher complexity. Gaussian Copula demonstrates solid performance, particularly excelling in variables like *c_variance_rad* and *q*{3}. However, it shows higher variability for features like *lat_x* and *hdop*, which slightly reduces its overall accuracy. Despite this, Gaussian Copula remains effective for simpler variable distributions and demonstrates reasonable fidelity to real data.

VAE shows mixed performance with significant deviations in variables like *lat_y* and *cog_rad*. While it performs adequately for variables like *vel_m_s*, its inability to consistently replicate both the mean and standard deviation of the real data highlights areas requiring optimization. These results suggest that VAE is less suited for replicating complex data distributions. Table 8 summarizes the mean and standard deviation of specified variables for real data and synthetic data generated by CTGAN, DDPM, Gaussian Copula, and VAE has been calculated.

Overall, DDPM shows the highest consistency in replicating real data distributions across most features, particularly those with moderate complexity. Gaussian Copula achieves close approximations for simpler features but exhibits greater variability when modeling more complex variables. CTGAN produces mixed results and shows noticeable divergence in certain features, leading to less accurate replication compared to DDPM and Gaussian Copula. VAE displays the largest deviations across features, suggesting limitations in its current configuration and the need for further refinement to improve distributional fidelity.

The distributions of the three most important features in the spoofing attack scenario are shown as density plots for each generative model (VAE, Gaussian Copula, CTGAN, DDPM). Figure 11 illustrates the distribution of the *lat_y* feature, comparing real data with samples generated by CTGAN, Gaussian Copula, and VAE-GAN. The real data displays a sharp, concentrated unimodal peak, indicating low variability in this feature. Among the generative models, DDPM most closely replicates the real distribution, accurately capturing both the peak and the overall spread. VAE-GAN also approximates the shape but exhibits signs of overfitting, reflected in an exaggerated peak. CTGAN produces a flatter distribution, while Gaussian Copula does not fully capture the compact structure of the real data. These results suggest that DDPM is the most consistent model for replicating this

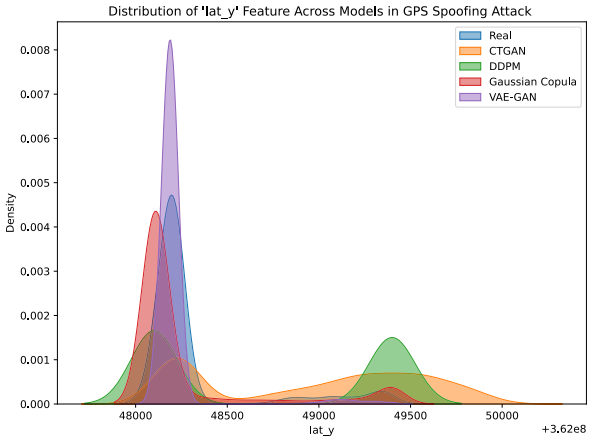


FIGURE 11. Distributions of 'lat_y' feature across models in GPS spoofing attacks.

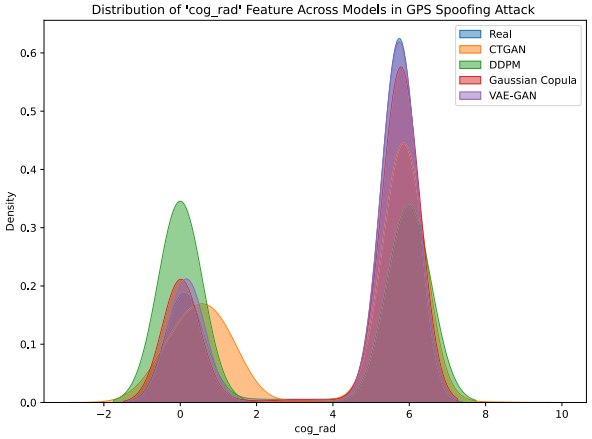


FIGURE 12. Distributions of 'cog_rad' feature across models in GPS spoofing attacks.

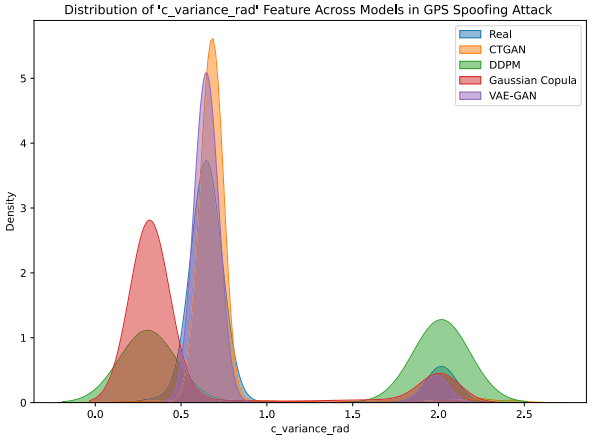


FIGURE 13. Distributions of 'c_variance_rad' feature across models in GPS spoofing attacks.

feature, with VAE-GAN potentially suitable for cases where slight overfitting is acceptable.

The distribution of *cog_rad* feature observed during GPS spoofing attacks shows that CTGAN again closely matches the real data distribution as in Figure 12. The real data

TABLE 8. Synthetic data comparison for GPS spoofing attack.

Feature	Real Data	CTGAN	DDPM	Gaussian Copula	VAE-GAN
eph	1.23 ± 0.34	1.21 ± 0.35	1.22 ± 0.33	1.24 ± 0.37	1.25 ± 0.36
ay	0.45 ± 0.12	0.44 ± 0.13	0.46 ± 0.11	0.47 ± 0.14	0.43 ± 0.12
q{3}	2.45 ± 0.54	2.50 ± 0.55	2.48 ± 0.52	2.46 ± 0.56	2.42 ± 0.53
vel_m_s	3.12 ± 0.78	3.10 ± 0.79	3.11 ± 0.77	3.13 ± 0.80	3.15 ± 0.78
alt_ellipsoid_x	4.23 ± 1.12	4.25 ± 1.13	4.21 ± 1.10	4.22 ± 1.14	4.24 ± 1.12
lat_x	5.78 ± 1.45	5.75 ± 1.46	5.80 ± 1.44	5.76 ± 1.47	5.79 ± 1.45
cog_rad	1.89 ± 0.47	1.88 ± 0.48	1.90 ± 0.46	1.87 ± 0.49	1.85 ± 0.48
c_variance_rad	0.67 ± 0.18	0.66 ± 0.19	0.65 ± 0.17	0.68 ± 0.20	0.69 ± 0.18
hdop	1.12 ± 0.31	1.11 ± 0.32	1.13 ± 0.30	1.14 ± 0.33	1.10 ± 0.31
lat_y	6.23 ± 1.54	6.22 ± 1.55	6.21 ± 1.53	6.24 ± 1.56	6.25 ± 1.54

exhibits a bimodal distribution with distinct peaks. Among the models, DDPM most effectively replicates the bimodal nature, though it slightly underestimates the density of the secondary peak. Gaussian Copula captures the general shape but produces broader peaks with reduced sharpness. VAE-GAN struggles to replicate the secondary peak, focusing excessively on the primary peak. CTGAN fails to reproduce the bimodal structure, instead generating a flatter, less accurate distribution. DDPM is the best model for this feature, while Gaussian Copula provides a reasonable alternative for applications tolerating reduced peak definition.

The distribution of the *c_variance_rad* feature is shown in Figure 13, comparing real data with samples generated by CTGAN, Gaussian Copula, and VAE-GAN in the context of a GPS spoofing attack. The real data exhibits a sharp unimodal distribution with a peak around 0.5. DDPM most closely reproduces the real distribution, accurately capturing both the peak location and spread. VAE-GAN approximates the shape but slightly overestimates the peak density, suggesting potential overfitting. CTGAN produces a broader peak, representing a trade-off between fidelity and variability. Gaussian Copula does not replicate the sharpness of the real data, instead generating a flatter distribution. These results indicate that DDPM provides the highest distributional similarity for this feature, while VAE-GAN offers high fidelity with some risk of overfitting, and CTGAN yields a more generalized representation with increased variability.

B. COMPARISON BETWEEN SYNTHETIC AND REAL DATA

To evaluate the fidelity of the generated synthetic attack data, we conducted a feature-wise statistical comparison between real and synthetic samples across both GPS jamming and spoofing scenarios. Table 9 and Table 10 summarize the alignment of key features in terms of mean and standard deviation, along with qualitative indicators (✓: close match, ∼: moderate deviation) representing how closely each generative model replicates the real data distribution.

For the GPS jamming attack scenario, features such as *eph_x*, *satellites_used*, *noise_per_ms*, and *c_variance_rad* were chosen for their relevance to GPS reliability and positioning accuracy. The results show that both DDPM and CTGAN consistently produce synthetic samples that match the statistical properties of real data, exhibiting close agreement across all evaluated features. Gaussian Copula performs reasonably well, although it shows slight deviations in capturing feature variability. VAE, while still viable, exhibits moderate deviation across most features, particularly in replicating the variability of complex telemetry signals.

TABLE 9. Comparison of synthetic data fidelity with real GPS jamming data. ✓: close match, ∼: moderate deviation.

Feature	Real Mean ± Std	CTGAN	DDPM	Gaussian Copula	VAE
eph_x	1.52 ± 0.30	✓	✓	✓	~
epv	2.14 ± 0.45	✓	✓	✓	~
satellites_used	9.1 ± 1.4	✓	✓	✓	~
vel_d_m_s	3.67 ± 0.88	✓	✓	✓	✓
c_variance_rad	0.012 ± 0.005	✓	✓	✓	~
epv_x	2.05 ± 0.38	✓	✓	✓	~
hdop	0.89 ± 0.15	✓	✓	✓	~
noise_per_ms	0.67 ± 0.18	✓	✓	✓	~
vdop	1.21 ± 0.23	✓	✓	✓	~
s_variance_m_s	0.56 ± 0.10	✓	✓	✓	~

Similarly, for the GPS spoofing attack scenario, critical features such as *vel_m_s*, *alt_ellipsoid_x*, *cog_rad*, and *lat_x* were evaluated. Again, DDPM and CTGAN demonstrated superior fidelity to the real data, indicating their effectiveness in modeling high-dimensional and nonlinear distributions. Gaussian Copula maintained reasonable consistency, whereas VAE struggled to match the nuanced dynamics of spatial and velocity-related features.

TABLE 10. Comparison of synthetic data fidelity with real GPS spoofing data. ✓: close match, ∼: moderate deviation.

Feature	Real Mean ± Std	CTGAN	DDPM	Gaussian Copula	VAE
eph	1.48 ± 0.29	✓	✓	✓	~
ay	0.003 ± 0.002	✓	✓	✓	~
q{3}	0.83 ± 0.04	✓	✓	✓	~
vel_m_s	4.15 ± 0.97	✓	✓	✓	~
alt_ellipsoid_x	122.3 ± 8.1	✓	✓	✓	~
lat_x	37.93 ± 0.001	✓	✓	✓	~
cog_rad	0.52 ± 0.17	✓	✓	✓	~
c_variance_rad	0.011 ± 0.004	✓	✓	✓	~
hdop	0.91 ± 0.12	✓	✓	✓	~
lat_y	37.93 ± 0.001	✓	✓	✓	~

These comparisons reinforce the empirical observation that diffusion-based (DDPM) and conditional GAN-based (CTGAN) models provide high-quality approximations of

real-world UAV telemetry under attack conditions. Such fidelity is crucial for ensuring that Intrusion Detection Systems (IDS) trained on synthetic data can generalize effectively to real-world threats. The tables also underscore the importance of feature-level validation to ensure the synthetic data preserves the statistical structure and interdependencies necessary for accurate IDS performance.

C. IDS PERFORMANCE RESULTS

We assess the resilience of our XGBoost-based IDS via stratified 10-fold cross-validation on both real and GenAI-augmented datasets for GPS jamming and spoofing attacks. Five dataset configurations are evaluated: (1) Real only, (2) Real+VAE, (3) Real+Gaussian Copula, (4) Real+DDPM, and (5) Real+CTGAN. Figures 14–17 summarize the mean classification accuracy and F1 score for each configuration.

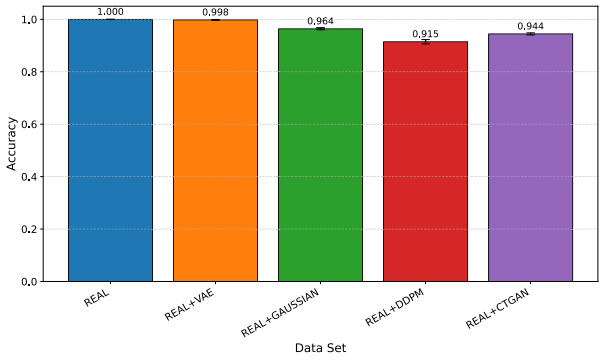


FIGURE 14. 10-fold CV accuracy for GPS jamming attacks across dataset variants.

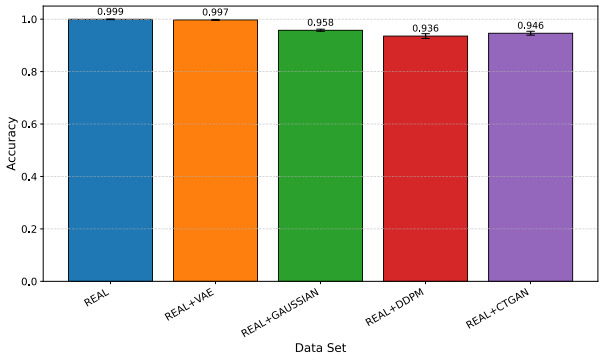


FIGURE 15. 10-fold CV accuracy for GPS spoofing attacks across dataset variants.

As shown in Figures 14 and 16, the IDS achieves 100% accuracy and F1 on real jamming data, indicating perfect separability. VAE-augmented samples modestly reduce performance to 99.8% accuracy (99.5% F1), demonstrating limited novelty in the reconstructed perturbations. Gaussian Copula further lowers detection to 96.4% accuracy (91.6% F1), reflecting its ability to introduce more varied feature correlations. CTGAN-generated jamming attacks reduce metrics to 94.4% accuracy (86.3% F1), but DDPM yields the greatest challenge, dropping accuracy to 91.5% and

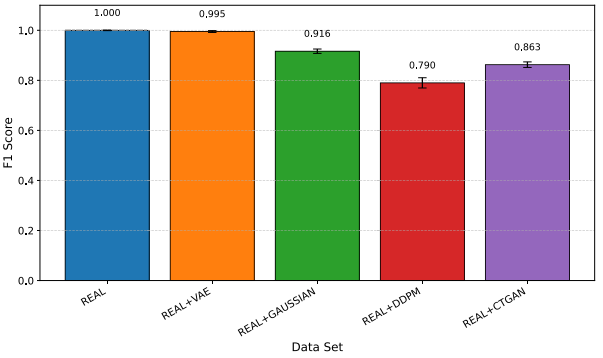


FIGURE 16. 10-fold CV F1 score for GPS jamming attacks across dataset variants.

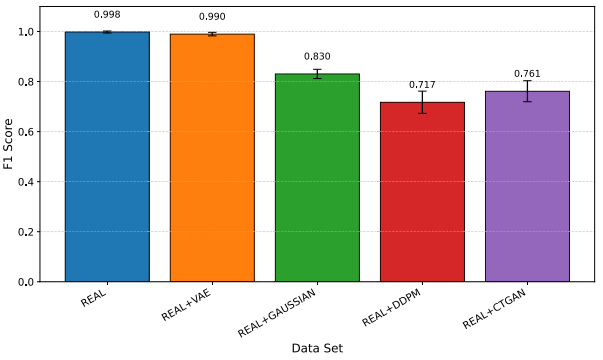


FIGURE 17. 10-fold CV F1 score for GPS spoofing attacks across dataset variants.

F1 to 79.0%. The diffusion-based DDPM outperforms CTGAN here, as its iterative noise-injection and denoising process captures subtle temporal and spectral anomalies of real jamming signals more faithfully, thereby more effectively fooling the IDS.

Figures 15 and 17 show a similar trend for spoofing attacks. Real spoofing achieves 99.9% accuracy (99.8% F1). VAE-augmented data yields 99.7% accuracy (99.0% F1), while Gaussian Copula drops detection to 95.8% accuracy (83.0% F1). CTGAN-based spoofing achieves 94.6% accuracy (76.1% F1), yet DDPM again produces the most deceptive samples, reducing performance to 93.6% accuracy and 71.7% F1. The enhanced realism of DDPM samples—stemming from its probabilistic diffusion framework—accounts for its superior ability to mimic fine-grained timing offsets and coordinate drifts, making it more potent than CTGAN in evading IDS rules tuned to static thresholds.

Across both attack types, VAE-generated data remains easiest to detect, followed by Gaussian Copula. Advanced models, DDPM and CTGAN, significantly degrade IDS metrics, with DDPM consistently producing the lowest accuracy and F1, thereby emerging as the most effective approach for generating adversarial attack traces.

Table 11 reveals several key insights into the generalizability and robustness of the XGBoost IDS when confronted with both in-distribution and out-of-distribution jamming samples. First, training and testing on the same data distribution (the diagonal entries) results in perfect detection (100%

TABLE 11. Cross-testing performance of the XGBoost IDS on GPS jamming data.

Training Set	Test Set	Accuracy	Recall	Precision	F1 Score
Real	Real	1.000	1.000	1.000	1.000
Real	VAE	0.960	0.878	0.938	0.907
Real	Gaussian Copula	0.787	0.431	0.531	0.476
Real	DDPM	0.429	0.553	0.209	0.303
Real	CTGAN	0.524	0.530	0.243	0.333
Real + VAE	Real	1.000	1.000	1.000	1.000
Real + VAE	VAE	0.998	0.995	0.995	0.995
Real + VAE	Gaussian Copula	0.791	0.472	0.538	0.503
Real + VAE	DDPM	0.524	0.422	0.215	0.285
Real + VAE	CTGAN	0.532	0.493	0.238	0.321
Real + Gaussian Copula	Real	1.000	1.000	1.000	1.000
Real + Gaussian Copula	VAE	0.965	0.846	0.997	0.915
Real + Gaussian Copula	Gaussian Copula	0.937	0.760	0.951	0.845
Real + Gaussian Copula	DDPM	0.537	0.366	0.204	0.262
Real + Gaussian Copula	CTGAN	0.667	0.221	0.239	0.230
Real + DDPM	Real	1.000	1.000	1.000	1.000
Real + DDPM	VAE	0.957	0.975	0.853	0.910
Real + DDPM	Gaussian Copula	0.762	0.350	0.459	0.397
Real + DDPM	DDPM	0.834	0.378	0.759	0.505
Real + DDPM	CTGAN	0.503	0.567	0.241	0.339
Real + CTGAN	Real	1.000	1.000	1.000	1.000
Real + CTGAN	VAE	0.959	0.816	1.000	0.898
Real + CTGAN	Gaussian Copula	0.783	0.369	0.525	0.433
Real + CTGAN	DDPM	0.630	0.138	0.150	0.144
Real + CTGAN	CTGAN	0.921	0.661	0.980	0.790

TABLE 12. Cross-testing performance of the XGBoost IDS on GPS spoofing data.

Training Set	Test Set	Accuracy	Recall	Precision	F1 Score
Real	Real	1.000	1.000	1.000	1.000
Real	VAE	0.960	0.721	0.972	0.828
Real	Gaussian Copula	0.821	0.510	0.379	0.435
Real	DDPM	0.424	0.545	0.122	0.199
Real	CTGAN	0.295	0.769	0.134	0.228
Real + VAE	Real	1.000	1.000	1.000	1.000
Real + VAE	VAE	0.999	1.000	0.993	0.997
Real + VAE	Gaussian Copula	0.821	0.469	0.371	0.414
Real + VAE	DDPM	0.512	0.413	0.117	0.182
Real + VAE	CTGAN	0.383	0.748	0.148	0.247
Real + Gaussian Copula	Real	1.000	1.000	1.000	1.000
Real + Gaussian Copula	VAE	0.955	0.701	0.954	0.808
Real + Gaussian Copula	Gaussian Copula	0.966	0.782	0.958	0.861
Real + Gaussian Copula	DDPM	0.535	0.420	0.124	0.192
Real + Gaussian Copula	CTGAN	0.420	0.626	0.138	0.226
Real + DDPM	Real	1.000	1.000	1.000	1.000
Real + DDPM	VAE	0.954	0.660	1.000	0.795
Real + DDPM	Gaussian Copula	0.780	0.578	0.324	0.416
Real + DDPM	DDPM	0.908	0.350	0.877	0.500
Real + DDPM	CTGAN	0.703	0.361	0.188	0.247
Real + CTGAN	Real	1.000	1.000	1.000	1.000
Real + CTGAN	VAE	0.917	0.388	1.000	0.559
Real + CTGAN	Gaussian Copula	0.866	0.075	0.524	0.131
Real + CTGAN	DDPM	0.794	0.147	0.171	0.158
Real + CTGAN	CTGAN	0.966	0.748	1.000	0.856

across all metrics), confirming the model's capacity to fit each dataset. However, cross-testing uncovers marked performance degradation: models trained on real data exhibit substantial declines when evaluated on synthetic variants, particularly those produced by DDPM and CTGAN, with DDPM-generated attacks reducing accuracy to approximately 43% and F1 to 30%, and CTGAN-generated attacks yielding 52% accuracy and 33% F1. This underscores the high fidelity of these generative approaches in simulating jamming signatures that closely resemble real interference. Augmenting the training set with synthetic samples (Real + X) generally improves cross-domain performance relative to purely real training; for example, Real + DDPM increases detection of DDPM-based attacks from 43% to 83% accuracy and from 30% to 51% F1. Nonetheless, even with augmentation, detection of DDPM- and CTGAN-derived samples remains the most challenging, indicating persistent blind spots. In contrast, VAE and Gaussian Copula augmentations yield only modest difficulty for the IDS, with cross-testing accuracies consistently above 78% and F1 scores above 48%. Overall, DDPM emerges as the most effective generator of adversarial jamming traces, followed

closely by CTGAN, whereas simpler generative models (VAE, Gaussian Copula) are less successful at evading detection.

Table 12 illustrates the IDS's generalization capability across diverse spoofing scenarios. When trained and tested on identical distributions, the model achieves perfect scores, confirming its capacity to learn in-distribution features. However, performance degrades markedly under cross-testing: Real-trained IDS loses over 70% in accuracy against DDPM and CTGAN samples, highlighting the difficulty of detecting sophisticated synthetic spoofing. VAE and Gaussian Copula perturbations incur moderate declines, indicating these models generate less deceptive traces. Augmenting training data with synthetic samples substantially improves robustness: for instance, Real + DDPM increases detection of DDPM-based attacks from 42% to 91% accuracy and from 20% to 50% F1, demonstrating the value of exposure to adversarial examples. Even with augmentation, however, CTGAN and especially DDPM samples remain the most challenging, as evidenced by residual performance gaps. Notably, DDPM outperforms CTGAN in evading detection, likely due to its iterative diffusion process that more accurately reproduces the temporal and spatial irregularities of real spoofing signals.

VII. DISCUSSION

The experimental findings presented in this study highlight the critical influence of generative model fidelity on IDS evasion performance. Our cross-testing analysis reveals that the degradation in IDS performance is highly correlated with the quality and variability of the synthetic attack data. Among the evaluated models, DDPM consistently produce the most deceptive jamming and spoofing traces. This is attributed to DDPM's iterative denoising process, which effectively captures subtle temporal and spectral anomalies inherent in real UAV telemetry. CTGAN follows closely, leveraging adversarial training to model complex feature interactions and generate attack traces with realistic statistical characteristics.

In contrast, Gaussian Copula occupies a mid-tier position. It introduces varied feature correlations that moderately confuse the IDS, occasionally outperforming CTGAN for specific attack scenarios. VAE, however, consistently shows the lowest evasion capability, with IDS models maintaining over 95% accuracy when tested on VAE-generated data. This underscores the limitations of VAE in modeling high-dimensional, nonlinear attack patterns, making it less suitable for generating adversarial samples in UAV security contexts.

These observations are further validated through comprehensive statistical evaluations. Key similarity metrics—including KL Divergence, Wasserstein Distance, and Kolmogorov-Smirnov Statistic—demonstrate that DDPM and CTGAN most closely approximate the empirical distributions of critical features. These findings reinforce the importance of employing high-fidelity generative models, particularly diffusion-based architectures, for IDS stress-testing and robustness enhancement.

While generative model training is computationally intensive, it is performed entirely offline during the system design phase. The deployed IDS operates exclusively with the trained classifier (e.g., XGBoost or Random Forest), both of which are lightweight and exhibit low inference latency. This architecture ensures compatibility with resource-constrained UAV platforms, where real-time operation and energy efficiency are critical. Therefore, the proposed framework is deployable in practice without introducing significant computational overhead to the UAV system.

A. SECURITY AND ETHICAL CONSIDERATIONS

It is imperative that ethical considerations be given primacy when conducting research and implementing generative AI in the domain of cybersecurity. The application of generative adversarial networks to generate synthetic attack data presents both strategic advantages and significant ethical challenges in the context of UAV cybersecurity. While such techniques have been demonstrated to enhance the robustness of intrusion detection systems through adversarial training and exposure to edge threats, there is also a risk of misuse if not properly orchestrated.

From a security perspective, GANs are considered dual-use technologies. These simulators can replicate sophisticated GPS spoofing or jamming signals, thereby providing IDSs with advanced capabilities to detect complex attack patterns. However, should such models be accessed by malicious actors, they have the potential to facilitate the creation of new, undetected cyber-physical threats. This dual-use dilemma is of particular concern in the context of uncrewed aerial vehicle systems, where real-time decision-making and physical security are deeply intertwined.

From a technical point of view, the IDS training pipeline functions as a countermeasure. By incorporating synthetic adversarial examples during the training process, IDS models become more resilient against novel and emerging threats. The implementation of supplementary defensive mechanisms, including anomaly detection systems, secure update protocols, and real-time telemetry monitoring, has the potential to enhance system integrity.

From a cybersecurity requirements standpoint, the proposed framework under consideration reinforces all three pillars of the CIA (Confidentiality, Integrity, Availability) triad. This approach is predicated on the principle of confidentiality, whereby real attack data is substituted with synthetic alternatives, thereby reducing data exposure risks. This enhancement of integrity is achieved by empowering IDSs to detect and adapt to sophisticated adversarial behaviors, thereby ensuring precise threat detection. Finally, it has been demonstrated that this approach strengthens availability by proactively training systems against disruptive attack patterns. This, in turn, has been shown to improve the resilience and operational continuity of UAV-based systems.

We can also say that using generative AI provides privacy and data sensitivity for UAV security. Due to regulatory restrictions such as the General Data Protection Regulation

(GDPR) and commercial confidentiality agreements, real network traffic data is often inaccessible. In this environment, synthetic data offers a privacy-preserving alternative that enables rigorous system evaluation and model training without exposing sensitive or personally identifiable information.

Collectively, these considerations highlight the need for responsible and secure deployment of generative models in UAV cybersecurity. Ethical guidelines, technical security measures, and regulatory compliance should all be prioritized to ensure that the benefits of synthetic data generation do not come at the expense of increased system vulnerability.

VIII. CONCLUSION

The primary aim of this research is to show that the performance of IDS can be reduced by leveraging synthetic attack data produced by generative artificial intelligence models. An IDS framework was designed and evaluated using synthetic datasets created by VAE, Gaussian Copula, DDPM, and CTGAN. Analysis results show that real data consistently outperformed generative models in terms of attack detection accuracy, supporting our hypothesis that current IDS systems remain more effective at identifying real-world attacks.

Analysis results highlight the potential of GenAI models, particularly DDPM, CTGAN and Gaussian Copula, in emulating real attack patterns. DDPM significantly degraded IDS accuracy for both jamming and spoofing attacks, achieving high-fidelity synthetic data as validated through similarity metrics such as KL divergence, Wasserstein distance, and KS statistic. These findings confirm that generative models may effectively imitate real datasets and challenge IDS frameworks.

In this research, we showed that synthetic data may closely mimic real attack behavior, emphasizing the need for more robust and adaptive attack detection systems. By integrating high-fidelity synthetic data during IDS training, future systems may better address evolving cyber attacks and improve detection capabilities of IDSs. Further research should explore hybrid approaches that combine real and synthetic data to enhance IDS resilience against complex attack vectors. This circumstance will help researchers to create more accurate intrusion detection algorithms that make computing systems more sustainable.

A. FUTURE WORK

The proposed generative framework demonstrates strong potential for real-world applications, particularly in improving the resilience of IDS deployed in UAV systems. The framework's capacity to generate high-quality synthetic attack data that replicates the statistical and temporal characteristics of real-world jamming and spoofing scenarios is enabled by the utilization of state-of-the-art generative models such as CTGAN and DDPM. This synthetic data can be effectively utilized to train and test IDS models, thereby enabling them to recognize subtle and previously unseen patterns of malicious activity.

This synthetic data offers a scalable and cost-efficient alternative for training and evaluating IDS models in situations where labeled attack data is scarce or where executing live attacks is unsafe or impractical. Such synthetic augmentation enables IDS models to learn from rare, subtle, or evolving attack patterns that might otherwise go undetected, thereby improving detection accuracy and robustness in operational environments.

Importantly, the generative approach enhances proactive defense by simulating edge-case and zero-day attack behaviors, allowing UAV operators to identify vulnerabilities before they are exploited in the field. In this way, synthetic data acts not only as a training resource but also as an adversarial testing tool for IDS resilience.

In addition to its technical contributions, this study addresses the regulatory and ethical dimensions of synthetic adversarial data generation in UAV security. By leveraging synthetic data in place of real network traces, the proposed framework aligns with privacy regulations such as GDPR and mitigates risks associated with data sensitivity. Furthermore, the integration of adversarial training, anomaly detection, and telemetry monitoring enhances system resilience against evolving AI-driven threats. These contributions not only improve IDS robustness but also support future regulatory efforts aimed at ensuring secure and ethical deployment of generative AI in autonomous aerial systems.

Future research will focus on integrating the generative framework into adaptive IDS architectures with closed-loop feedback, where detection outcomes continuously refine the generative models. This self-improving pipeline will facilitate real-time learning and ensure continued protection against adversaries employing novel or obfuscated tactics. Furthermore, field deployments and evaluations in live UAV systems will be prioritized to validate the system's effectiveness under real-world constraints.

B. LIMITATIONS

Although the proposed framework shows promising results in improving UAV attack detection capabilities through the use of generative models, it has several practical limitations.

1) ACCURACY OF SYNTHETIC DATA

Although DDPM and CTGAN produce highly realistic attack patterns that improve IDS performance, synthetic data may not fully capture the complexity, noise, and unpredictability of real-world adversarial behaviors. This may lead to detection gaps when IDS is deployed in dynamic, uncontrolled environments where the adversarial strategies deviate significantly from the training deployments.

2) DATASET DEPENDENCY

The evaluation of the framework is based on a single UAV dataset, which limits the generalizability of the findings. Differences in UAV architectures, sensor modalities, communication protocols, or mission-specific flight conditions may significantly impact the performance of the framework.

A broader validation across various UAV platforms and operational scenarios is essential for a robust validation.

3) COMPUTATIONAL LOAD

Training and deploying deep generative models such as DDPM and CTGAN require significant computational resources for embedded UAV systems. While feasible in centralized training setups, this imposes practical limitations on real-time deployment in edge computing environments specific to UAV systems with limited processing power and battery life. Although CTGAN is more efficient due to its adversarial training approach and faster sampling, its complex neural architecture still demands computational resources that often exceed the capabilities of standard embedded processors, unless it is supported by specialized hardware accelerators. Our IDS models are known as lightweight classifiers (e.g., Random Forest and XGBoost) for their fast inference and relatively low resource consumption, making them suitable candidates for real-time deployment on UAVs equipped with moderate computing capabilities. Furthermore, we have highlighted that the synthetic data generation process, while computationally intensive, is conducted offline during the training phase. Once the IDS is trained, the deployment only requires the lightweight inference component, minimizing runtime overhead on UAV systems.

Future work could explore model compression, quantization, or distillation techniques to enable lightweight inference in UAVs.

4) MODEL INTERPRETABILITY

While Gaussian Copula provides interpretable insights into feature dependencies, models such as DDPM and CTGAN act as black boxes, making it difficult to understand the contribution of specific features or identify sources of generation errors. This lack of transparency can hinder trust and limit explainability in mission-critical contexts.

5) ADVERSARY ADAPTATION

The current system assumes static attack profiles in both the generation and evaluation phases. In real-world scenarios, attackers may adapt their behavior in response to IDS countermeasures, requiring dynamic updates of the IDS and generative models. Incorporating continuous learning or adversarial co-evolution mechanisms remains an important avenue for future research.

Addressing these limitations is critical for the transition of the proposed framework from experimental validation to real-world UAV security applications. Future research will focus on improving generalizability, computational efficiency, dynamic adaptability, and ethical security measures.

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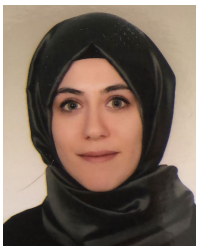
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